CHAPTER 9

Boson Expansion Methods

9.1 Introduction

In the last chapter we studied collective vibrations within the harmonic approximation. Harmonic oscillations are characterized by an equidistant spectrum. As shown in Fig. 1.5, in the quadrupole case we expect a one-boson 2^+ excitation at energy $\hbar\omega$, a two-boson triplet 0^+, 2^+, 4^+ at twice this energy, a three-boson quintet 0^+, 2^+, 3^+, 4^+, 6^+ at three times this energy, and so on. Experimental spectra of spherical nuclei (see, for instance, Fig. 9.4) in some regions of the periodic table qualitatively show this structure. In detail, however, we always find some more or less dramatic deviations from this simple picture, as, for instance, splitting of the higher multiplets and a shift in the position of the energy centroid.

Such anharmonicities are caused by two effects:

(i) The collective fermion pairs*

$$\sum_{mi} C_{mi} a_m^+ a_i \tag{9.1}$$

are not exact bosons. Applying this operator twice, in the description of a two-phonon state, we do not obtain two uncoupled bosons,

* For the sake of simplicity, in this chapter we investigate mainly bosons based on TDA wave functions. In principle, we could also use RPA bosons of the form (8.65). This has not been done very often in the literature [Sø 70], because in the method of boson expansions higher correlations are taken into account anyway.

because the Pauli principle excludes all configurations, where the same level is occupied by more than one particle (or hole). We call such effects *kinematic effects* because they will be connected with the basis used in the following sections.

(ii) The exact many-body Hamiltonian contains not only second-order terms in the bosons, which are diagonalized in the TDA approach (or the RPA method), but also higher-order terms. Such terms are neglected in the harmonic approximation and provide a coupling between the different collective modes and also a coupling to non-collective states, such as rather pure two-quasi-particle states. We call such effects *dynamic effects*, because they are determined by the properties of the Hamiltonian.

If we start with a double magic nucleus with a rather pure harmonic spectrum and add more and more particles, we observe larger and larger *anharmonicities*. We come into a transitional region where the vibrational character of the spectrum is lost and a rotational structure develops. Finally, we end up in the region of well-deformed nuclei with pure rotational bands.

We are able to describe such anharmonicities and, in principle, also the transition to deformed nuclei by the methods discussed in this chapter. The basic idea of such methods is to represent the fermion Hamiltonian by pure boson operators B_μ, B_μ^+ and to diagonalize it in a boson space. Mathematically, this corresponds to a mapping of the Fock space of many fermion states into a space of boson states. This will be discussed in detail in Section 9.2.

The advantage of such a boson representation is that very collective modes—for instance, the quadrupole mode—can be approximated by rather simple wave functions, namely one-boson states $B_\mu^+|0\rangle$. For such states the quadratic approximation of the Hamiltonian is already very good, as we saw in the last chapter. We can therefore hope that higher terms in the boson representation of the Hamiltonian drop off rapidly. In Section 9.2 we shall see that this is, in fact, the case for collective modes. It is often sufficient to take into account only fourth- or sixth-order terms.

On the other hand, pure *ph* operators $a_m^+ a_i$ which are needed for the description of non-collective modes, have a complicated structure in the boson space. The basic idea of all boson expansions is therefore to restrict ourselves to a few collective bosons which span a *collective subspace* of the full Hilbert space, and to diagonalize the Hamiltonian within this subspace. If there are eigenstates of the exact Hamiltonian in this subspace, that is, if we can neglect their coupling to non-collective states in the Hilbert space, we obtain a rather simple description of these states.

This assumption is certainly not fulfilled for all states or for all nuclei. In many problems we must explicitly take into account the coupling to non-collective few-quasi-particle states. This is in particular always the case for odd systems. In Section 9.3 we will therefore treat techniques that

deal with the coupling of collective modes (described by the boson picture) to single-particle states. One of the simplest models of this kind is the *vibration particle coupling* (VPC) model of Bohr and Mottelson.

9.2 Boson Representations in Even–Even Nuclei

9.2.1 Boson Representations of the Angular Momentum Operators

The problem of representing a fermion system by Boson operators has been investigated in many fields of physics (for a review see [Ga 78]). Since such representations are in general highly mathematical, in this section we shall start with some simple examples that already show the basic features of such methods.

The *angular momentum operators* can be represented by bosons. In the theory of spin waves in a ferromagnet, several techniques of this type have been developed, and it has turned out that some of them can be extended to the case of general many-fermion systems.

We shall start with the algebra of the angular momentum operators $J_{\pm}=J_x \pm iJ_y$ and J_z, which obey the commutation relations of the group $SU(2)$:

$$[J_+, J_-]=2J_z, \qquad [J_z, J_+]=J_+. \tag{9.2}$$

The corresponding Hilbert space is spanned by the eigenvectors $|IM\rangle$ of $\mathbf{J}^2$ and J_z. This Hilbert space is now mapped onto a space of bosons, given by the boson vacuum* $|0)$ and boson operators B, B^+, which fulfill the relations

$$[B, B^+]=1, \qquad B|0)=0. \tag{9.3}$$

There are several ways to carry out this mapping, that is, to represent the operators J_i by the boson operators B, B^+.

(i) The Holstein Primakoff Representation [HP 40]. Restricting ourselves to the subspace characterized by the angular momentum quantum number I, we can map the operators $J_{\pm}$ and J_z in the following way.

$$\begin{aligned} J_+ &\to (J_+)_B=\sqrt{2I}\cdot B^+\sqrt{1-\frac{1}{2I}\cdot B^+B}\ , \\ J_- &\to (J_-)_B=(J_+)_B^+=\sqrt{2I}\cdot\sqrt{1-\frac{1}{2I}B^+B}\cdot B, \\ J_z &\to (J_z)_B=-I+B^+B. \end{aligned} \tag{9.4}$$

*In the following we always use sharp brackets $|\ \rangle$ for the original angular momentum and fermion states, and round brackets $|\ )$ for the states in the boson space.

The square roots are formal abbreviations for the corresponding Taylor series. It is easy to show that the operators $(J_{\pm})_B$ and $(J_z)_B$ in the boson space satisfy the angular momentum algebra (9.2).

The boson space is spanned by the n-boson states

$$|n)=\frac{1}{\sqrt{n!}}(B^+)^n|0) \qquad \text{for } n=0,1,2,\ldots. \tag{9.5}$$

In contrast to the subspace with angular momentum I, which has the dimension $2I+1$, the boson space may have infinite dimension ($n\to\infty$). On the other hand, because B^+B counts the number of bosons (n), the square roots in Eq. (9.4) are not defined in the whole boson space. Thus we have the condition

$$n\leqslant 2I. \tag{9.6}$$

The subspace which, at maximum, contains $2I$ bosons is called the *physical subspace* of the infinitely dimensional boson space. It is spanned by the $2I+1$ states

$$|0), B^+|0),\ldots,(B^+)^{2I}|0) \tag{9.7}$$

and has therefore the same dimension as the angular momentum space. The mapping (9.4) provides a one-to-one correspondence between the angular momentum space characterized by I and the physical subspace. Since it conserves the commutation relations, it is a matter of convenience in which space we want to carry out actual calculations. All physically important properties are not influenced by this mapping.

This is the basic idea of all boson representations, and we already see the two basic ingredients that have to be fulfilled for this concept:

(i) We have to be sure that the algebraic rules for calculations, that is, the underlying *algebra*, are conserved by the mapping

(ii) One is only allowed to work in the physical subspace, i.e., in our example all the vectors have to belong to the eigenspace of the operator $\mathbf{J}^2$ with the eigenvalue $I\cdot(I+1)$. In general one can express this fact in the following way: One can only work with vectors, which are eigenstates to the *Casimir operators* of the underlying symmetry group [Ya 74]

As an example, we express the eigenvectors of the angular momentum states by boson states:

$$|IM\rangle\to|IM)=\left[(I+M)!\right]^{-1/2}(B^+)^{I+M}|0). \tag{9.8}$$

In practical applications we have to use the Taylor expansion of Eq. (9.4). It converges well, if the number of bosons is small compared to $2I$. It is a disadvantage of this type of boson representation that, in principle, an infinite number of terms is needed to fulfill the commutator algebra (9.2). There are, however, other ways to avoid this difficulty:

(ii) The Dyson Representation [Dy 56]. Dyson proposed the following mapping.

$$\begin{aligned} J_+ &\to (J_+)_B = \sqrt{2I}\, B^+\left(1-\frac{1}{2I}B^+B\right), \\ J_- &\to (J_-)_B = \sqrt{2I}\, B, \\ J_z &\to (J_z)_B = -I + B^+B. \end{aligned} \tag{9.9}$$

This representation is very simple, but it violates Hermiticity because $(J_+)_B^+$ is different from $(J_-)_B$, and leads to a non-Hermitian Hamiltonian. However, as we shall see in the following sections, it has certain advantages and can also be used in the many-body problem.

(iii) The Schwinger Representation [Sch 65]. Schwinger introduced a boson representation of the angular momentum operators which, as well as being finite, is also Hermitian. He used two bosons A and B,

$$[A, A^+]=1, \qquad [B, B^+]=1, \qquad [A, B]=[A, B^+]=0, \tag{9.10}$$

which act in a space of two-dimensional oscillator states

$$|n_A n_B) = \frac{1}{\sqrt{n_A!\, n_B!}}(A^+)^{n_A}(B^+)^{n_B}|0,0). \tag{9.11}$$

The Schwinger representation is given by

$$\begin{aligned} J_+ &\to (J_+)_B = B^+A, \\ J_- &\to (J_-)_B = (J_+)_B^+ = A^+B, \\ J_z &\to (J_z)_B = \tfrac{1}{2}(B^+B - A^+A). \end{aligned} \tag{9.12}$$

The eigenstates of the angular momentum operators can be written as

$$|IM) = [(I-M)!(I+M)!]^{-1/2}(B^+)^{I+M}(A^+)^{I-M}|0,0). \tag{9.13}$$

The boson number operator

$$N = B^+B + A^+A = 2I \tag{9.14}$$

is proportional to I. Rotational invariant Hamiltonians must therefore conserve the boson number. This is different in the Holstein–Primakoff representation, which yields boson number violating terms. On the other hand, it can be shown that there is a nearly unitary transformation between both representations [BM 78].

It is evident that these simple representations of the angular momentum operators can be used in *models*, in which the Hamiltonian is expressed by spin or quasi-spin operators, like the Lipkin model (see Sec. 5.4) or the seniority model (see Sec. 6.2). In such cases, we immediately obtain a boson representation of the Hamiltonian. Such models have therefore been widely used to study the properties of boson representations [PKD 68, BS 68b, Kl 69], and we will come back to some of them in the following sections.

Such boson representations of the angular momentum operators have also been used in the *rotor model* (see Sec. 1.5). Marshalek [Ma 75b] introduced a generalization of Eq. (9.4) with three bosons corresponding to the three degrees of freedom usually described by the Euler angles. Yamamura et al. [YSI 78, IY 78] used an extension of the Schwinger representation with four bosons.

These methods are most useful in the limit of larger angular momenta, where there is essentially a classical rotation around a fixed axis. In a boson picture, deviations from this rotation may be described as small oscillations of the rotational axis, so-called wobbling bosons [BM 75, Ma 79].

9.2.2 Concepts for Boson Expansions

In the last section we discussed several examples of boson representations of the angular momentum operators. These methods can be generalized to the case of a many-fermion system. We shall first restrict ourselves to systems with *even particle number*, systems with odd particle number will be described in Section 9.3.

As in the angular momentum case, there are many types of boson representations. Some of them are equivalent in certain limits. Before we go into the details, we want to discuss a few basic concepts which so far have been used in the theory of boson representations.

All these methods start with a many-fermion Hilbert space $\mathfrak{H}_F$, which contains vectors and operators. In second quantazation the space is completely specified by the vacuum $|0\rangle$ and a set of fermion operators α_k, α_k^+. In the following we always work in the quasi-particle picture, that is, the operators are the Bogoliubov quasi-particle operators in the sense of Chapter 7; the HF-quasi-particles are just a special case of them.

Since bosons have integer spin values we prefer to use pairs of fermion operators

$$\alpha_k^+ \alpha_l^+, \qquad \alpha_l \alpha_k, \qquad \alpha_l^+ \alpha_k. \tag{9.15}$$

In fact, we can describe all wave functions and operators in an even system by such fermion pairs and the corresponding quasi-particle vacuum $|0\rangle$.*

This fermion space is now mapped onto a different Hilbert space, a space of bosons $\mathfrak{H}_B$

$$\mathfrak{H}_F \rightarrow \mathfrak{H}_B. \tag{9.16}$$

* If we start with a vacuum $|0\rangle$ corresponding to an odd system, we are, in principle, also able to describe odd systems completely by the pair operators (9.15). Single-fermion operators are only needed for transfer processes. For practical reasons (non-degenerate vacuum), however, we prefer to use only quasi-particle vacua with even particle number.

The boson space is given by a boson vacuum $|0)$ and boson creation and annihilation operators* B_μ, B_μ^+:

$$B_\mu|0)=0, \qquad \left[B_\mu, B_{\mu'}^+\right]=\delta_{\mu\mu'}, \qquad \left[B_\mu, B_{\mu'}\right]=0. \tag{9.17}$$

As in the angular momentum case (Sec. 9.2.1), it generally turns out that the boson space $\mathfrak{H}_B$ is much larger than the fermion space $\mathfrak{H}_F$. Therefore, the mapping (9.16) is only unique in a certain subspace $\mathfrak{H}_{\text{phys}}$ of the boson space

$$\mathfrak{H}_F \leftrightarrow \mathfrak{H}_{\text{phys}} \subset \mathfrak{H}_B.$$

There are now two basic concepts to introduce this mapping† explicitly:

(i) Belyaev and Zelevinskii [BZ 62] propose to map the operators in such a way that the *commutation relations* are preserved. Usually all important operators can be constructed from a set of *basic operators* whose commutation relations form an *algebra*. An example is the set (9.15) and another example is the operators $K_\pm$ and K_0 in the case of the Lipkin model (Sec. 5.4). It is then sufficient to map only these basic operators onto operators in the boson space in such a way that this algebra is preserved.‡ All other operators can be represented as polynomials of these basic operators and their boson image is obtained as the same polynomial in the images of the basic operators.

To obtain the mapping of the vectors one only has to define a correspondence between the vacuum in the boson space $|0)$ and the quasiparticle vacuum $|0\rangle$ in the fermion space. All other states can be found by successive application of the basic operators.

It is evident that the vectors obtained in this way do not span the whole boson space but only the physical subspace. It is an eigenspace to the Casimir operators of the corresponding commutator algebra, as discussed in Section 9.1. In cases where the commutator algebra is fulfilled only approximatively, we can have admixtures of unphysical states. To avoid them, we have to ensure at least that one works in eigenspaces of the Casimir operators [Ya 74].

(ii) Marumori [MYT 64a and b] proposed to *map vectors* in the Hilbert spaces $\mathfrak{H}_F$ and $\mathfrak{H}_B$ and to define the operators in such a way that the

*In the following we shall always use sharp brackets $|\ \rangle$ for states in the fermion space $\mathfrak{H}_F$ and round brackets $|\ )$ for states in the boson space $\mathfrak{H}_B$. We also use small Greek letters for fermion operators and capital Latin letters for bosons. The fermion indices are given by Latin letters $k, l, \ldots$; the boson indices are Greek letters $\mu, \nu, \ldots$. Sometimes a boson index corresponds to a fermion pair (k, l).

†We also can use the generator coordinate method to introduce boson representations [JS 64, JDF 71, Ho 72, HL 72b]. We will study this technique in more detail in Section 10.4.

‡In the case of the Lipkin model, for instance, this is achieved by one of the prescriptions discussed in Section 9.2.1.

matrix elements are conserved by the mapping. In practice, one starts with a full orthogonal set of basis states $|n)$ in the boson space:

$$(n|n') = \delta_{nn'}; \qquad \sum_n |n)(n| = 1_B$$

and maps them onto states $|n\rangle$ in the Fermion space.

The image of an operator O_F in the fermion space is then given by

$$O_F \to O_B = \sum_{nn'} \langle n|O_F|n'\rangle \cdot |n)(n'|. \tag{9.18}$$

This definition obviously conserves the matrix elements. The mapping of Fermion states $|n\rangle$ to boson states by this prescription, however, is not unique. In Section 9.2.6 we will see in some detail how this mapping can be carried out in a unique manner by restricting ourselves to the physical subspace.

These two concepts of mapping *operators* (Belyaev–Zelevinskii) or *states* (Marumori) have been applied in different ways. Before we discuss some of the methods (in the following sections), we wish to stress that in all the applications we must, to a greater or lesser extent, use drastic approximations, which are justified only in special physical cases. For instance, an expansion of Eq. (9.4) in a Taylor series with only a few terms is only meaningful for large values of I, that is, in a case where J_+ already has well-pronounced boson properties. It is therefore important to choose the basic fermion operators that are to be mapped in a first approximation onto a boson in a proper way, such that rapid convergence is guaranteed. Mathematically it is possible to start with pure two-quasi-particle operators $\alpha_k^+ \alpha_l^+$ (see Sec. 9.2.6). Since they only fulfill the boson commutation relations to a very poor approximation, the corresponding boson expansion does not converge rapidly and is useless in practical applications.

Boson representations are therefore only meaningful if we start with linear combinations of fermion pairs

$$b_\mu^+ = \sum_{k<l} C_{kl}^\mu \alpha_k^+ \alpha_l^+, \tag{9.19}$$

which in the following we call *collective fermion pair operators*.* The coefficients $C_{kl}^\mu = -C_{lk}^\mu$ can be obtained, for instance, from TDA calculations (see Sec. 8.2), or we can use operators of a given type, such as the five quadrupole operators

$$b_\mu^+ = \frac{1}{2} \sum_{kl} Q_{kl}^{2\mu} \alpha_k^+ \alpha_l^+ \qquad (\mu = -2, \ldots +2). \tag{9.20}$$

* For a discussion of the term "collective," see Section 8.2.2.

They are easy to handle because of their group theoretical properties, and have been widely used in the literature. We have to be aware, however, that whenever we restrict the number of given operators b_μ^+, the operators (9.19) no longer form a complete system in the two-quasi-particle space, and we fail to describe certain states not taken into account. For instance, it is impossible to approximate a pure two-quasi-particle state in the space spanned by pure quadrupole bosons.

So far we have considered the collective low-lying excitations of a many-fermion system as approximate bosons and used them to construct a theory based on the corresponding ideal bosons. A different system, in which we have had to deal with approximate bosons, is that of the superfluid system of Cooper pairs studied in Section 6.2. In this case, we do not start with an HF ground state as quasi-particle vacuum and *ph* excitation as quasi-bosons. We start with the bare vacuum and couple pairs of particles to the quasi-bosons. As we shall see in Section 9.2.7, such an approach is closely connected to the idea of the *Schwinger representation* [see Eq. (9.12)]. It has the advantage that symmetries, such as the particle number conservation, are guaranteed from the beginning, but also has the disadvantage that we have to construct the ground state of the system out of many bosons, which might be a very complicated task in the general case.

After these general remarks, in the following sections we shall use bosons based on collective *ph* (or two-quasi-particle states) and discuss in more detail a few methods used to describe the many-fermion system in this way.

9.2.3 The Boson Expansion of Belyaev and Zelevinskii

We begin our discussion with a quasi-particle representation of the many-fermion space. It is based on a generalized Slater determinant $|0\rangle$, which represents the vacuum of the quasi-particle operators $\{\alpha_1 \ldots \alpha_M\}$. The Hamiltonian has the form* (E.18)

$$H = H^0 + H^{11} + H^{22} + H^{31} + H^{40}. \tag{9.21}$$

In an even system it is sufficient to consider only the fermion pair operators†

$$b_{kl}^+ = \alpha_k^+ \alpha_l^+; \qquad b_{kl} = \alpha_l \alpha_k; \qquad a_{kl} = \alpha_l^+ \alpha_k. \tag{9.22}$$

* H^{20} does not occur, since we assume that the ground state $|0\rangle$ is a self-consistent solution of the variational principle (7.31). This fact is, however, not essential for the following discussions.

† In the following we will always use small Latin letters for fermion pair operators and capital Latin letters for bosons. They take the pair index (l, k) in the non-collective case and the Greek indices $\mu, \nu, \ldots$ in the collective case.

They form an algebra [which belongs to the group $SO(2M)$]:

$$[b_{kl}, b_{k'l'}] = 0, \tag{9.23a}$$

$$[b_{kl}, b^+_{k'l'}] = \delta_{kk'}\delta_{ll'} + \delta_{kl'}a_{lk'} + \delta_{lk'}a_{kl'} - (k \leftrightarrow l), \tag{9.23b}$$

$$[b_{kl}, a_{k'l'}] = \delta_{ll'}b_{kk'} - (k \leftrightarrow l), \tag{9.23c}$$

$$[a_{kl}, a_{k'l'}] = \delta_{kl'}a_{k'l} - \delta_{lk'}a_{kl'}. \tag{9.23d}$$

In principle, non-collective fermion pair operators can be represented by corresponding bosons, and we will show how to do this in more detail in Section 9.2.6. However, as discussed in the last section, for all practical applications we need collective bosons. We therefore start with the collective fermion pair operators

$$b^+_\mu := \sum_{k<l} C^\mu_{kl}\alpha^+_k\alpha^+_l, \tag{9.24}$$

$$a_{\mu\nu} := \sum_{klr} C^{\mu*}_{kr}C^\nu_{lr}a_{kl} = a^+_{\nu\mu}. \tag{9.25}$$

The coefficients $C^\mu_{kl} = -C^\mu_{lk}$ characterize the kind of bosons we will use for our representation. We shall assume that they obey the orthogonality and completeness relations (8.26)

$$\sum_{k<l} C^{\mu*}_{kl}C^{\mu'}_{kl} = \delta_{\mu\mu'}; \qquad \sum_\mu C^\mu_{kl}C^{\mu*}_{k'l'} = \delta_{kk'}\delta_{ll'} - (k \leftrightarrow l). \tag{9.26}$$

As long as we take into account all μ-values, Eq. (9.24) represents a unitary transformation in the space of fermi pair operators. In practical applications we often use only one or a few bosons, which means that care has to be taken in using the completeness relations (9.26).

We can now use Eq. (9.24) to derive the commutator algebra for the collective fermion pair operators (9.24) and (9.25):

$$[b_\mu, b_{\mu'}] = [b^+_\mu, b^+_{\mu'}] = 0, \tag{9.27a}$$

$$[b_\mu, b^+_{\mu'}] = \delta_{\mu\mu'} - a_{\mu\mu'}, \tag{9.27b}$$

$$[b_\mu, a_{\mu'\nu'}] = \sum_\rho \Gamma^{\mu\mu'}_{\nu'\rho}b_\rho, \tag{9.27c}$$

$$[b^+_\mu, a_{\mu'\nu'}] = -\sum_\rho \Gamma^{\mu\nu'*}_{\mu'\rho}b^+_\rho, \tag{9.27d}$$

$$[a_{\mu\nu}, a_{\mu'\nu'}] = \frac{1}{M-2}\sum_{\rho\sigma}\left(\Gamma^{\mu'\mu\sigma}_{\nu\nu'\rho} - \Gamma^{\mu\mu'\sigma}_{\nu'\nu\rho}\right)a_{\rho\sigma}, \tag{9.27e}$$

where M is the dimension of the single-particle configuration space. (This factor drops out in the final formulae, so there need be no worry about an infinite space.) The coefficients $\Gamma^{\mu\nu}_{\mu'\nu'}$, $\Gamma^{\mu\nu\rho}_{\mu'\nu'\rho'}$, and other quantities $\Gamma^{\mu k}_{\mu' k'}$,

which we shall need later on, are defined by

$$\Gamma^{\mu\ \nu}_{\mu'\nu'} = \sum_{klpq} C^{\mu^*}_{kl} C^{\mu'}_{lp} C^{\nu^*}_{pq} C^{\nu'}_{qk}, \tag{9.28a}$$

$$\Gamma^{\mu\ \nu\ \rho}_{\mu'\nu'\rho'} = \sum_{klpqrs} C^{\mu^*}_{kl} C^{\nu'}_{lp} C^{\rho^*}_{pq} C^{\rho'}_{qr} C^{\nu^*}_{rs} C^{\mu'}_{sk}, \tag{9.28b}$$

$$\Gamma^{\mu\ \nu\ \rho\ \sigma}_{\mu'\nu'\rho'\sigma'} = \cdots$$

$$\Gamma^{\mu\ k}_{\mu' k'} = \sum_{l} C^{\mu^*}_{k'l} C^{\mu'}_{kl}. \tag{9.29}$$

Figure 9.1 shows an obvious graphical representation of Eqs. (9.28). The quantities Γ are closely related to the fact that the fermion pair operators b_μ are not real bosons. They take into account the Pauli principle by taking care of exchange corrections. Using Eqs. (9.26), we can prove many useful relations, as, for example,

$$\begin{aligned}
&\sum_{\tau} \Gamma^{\tau\mu\mu'\cdots}_{\tau\nu\nu'\cdots} = -(M-1)\Gamma^{\mu\mu'\cdots}_{\nu\nu'\cdots}, \qquad \Gamma^{\mu}_{\nu} = -2\delta_{\mu\nu};\\
&\sum_{\tau} \Gamma^{\mu\ \nu}_{\mu'\tau} \Gamma^{\tau\rho}_{\nu'\rho'} = -\Gamma^{\mu\ \nu\ \rho}_{\mu'\nu'\rho'} - \Gamma^{\mu\ \nu\ \rho}_{\mu'\rho'\nu'};\\
&\sum_{\sigma\rho} \Gamma^{\mu\sigma}_{\nu\rho} \Gamma^{\rho\lambda}_{\sigma\tau} = (M-2)\Gamma^{\mu\lambda}_{\nu\tau} + 4\delta_{\mu\nu}\delta_{\lambda\tau};\\
&\sum_{\sigma\rho} \Gamma^{\rho l}_{\sigma k} \Gamma^{\sigma p}_{\rho q} = (M-2)\delta_{pk}\delta_{ql} + \delta_{pq}\delta_{kl}.
\end{aligned} \tag{9.30}$$

To get Eq. (9.27), we need the inverse of (9.25), which reads

$$a_{kl} = \frac{1}{M-2}\left\{ \sum_{\mu\nu} \Gamma^{\nu k}_{\mu l} a_{\mu\nu} - \delta_{kl} \frac{1}{M-1} \sum_{\mu} a_{\mu\mu} \right\}, \tag{9.31}$$

and can be verified by a simple insertion of Eq. (9.25). We can also represent the Hamiltonian (E.18) in terms of the collective fermion pair

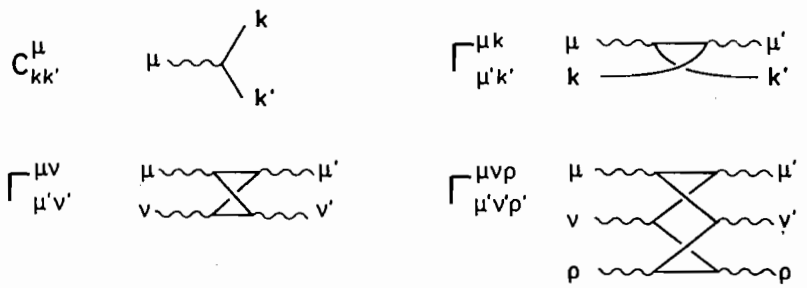


Figure 9.1. Graphical representation of the quantities C and Γ. (These kinds of graphs are not Feynman graphs in the sense of Appendix F.)

operators b_μ and $a_{\mu\nu}$:

$$\begin{aligned} H^{11} &= \sum_{\mu\nu} E_{\mu\nu} a_{\mu\nu}, \\ H^{22} &= \sum_{\mu\nu} V^{22}_{\mu\nu} b^+_\mu b_\nu, \\ H^{31} &= \sum_{\mu\nu\rho} V^{31}_{\mu\nu\rho} b^+_\mu a_{\nu\rho} + \text{h.c.}, \\ H^{40} &= \sum_{\mu\nu} V^{40}_{\mu\nu} b^+_\mu b^+_\nu + \text{h.c.}, \end{aligned} \tag{9.32}$$

with

$$\begin{aligned} E_{\mu\nu} &= \frac{1}{M-2} \sum_k \left(\Gamma^{\nu k}_{\mu k} - \frac{1}{M-1} \delta_{\mu\nu} \right) E_k, \\ V^{22}_{\mu\nu} &= \frac{1}{4} \sum_{klpq} H^{22}_{klpq} C^{\mu^*}_{kl} C^{\nu}_{pq}, \\ V^{31}_{\mu\nu\rho} &= \frac{1}{M-2} \sum_{klpq} H^{31}_{klpq} C^{\mu^*}_{kl} \left(\Gamma^{\rho q}_{\nu p} - \frac{1}{M-1} \delta_{\rho\nu} \delta_{pq} \right), \\ V^{40}_{\mu\nu} &= \sum_{klpq} H^{40}_{klpq} C^{\mu^*}_{kl} C^{\nu^*}_{pq}. \end{aligned} \tag{9.33}$$

A special case is the *Lipkin model* (see Sec. 5.4), which we shall discuss later on in some detail. In the following, we apply this model using the quasi-particle representation, in which the operator $\alpha^+_m = c^+_m$ (and $\beta^+_m = c^+_{-m}$) correspond to the creation of a particle in the upper shell (and a hole in the lower shell). In the HF state $|0\rangle$ the lower shell (Ω particles) is completely filled

$$\alpha_m |0\rangle = \beta_m |0\rangle = 0. \tag{9.34}$$

The Hamiltonian commutes with the quasi-spin operator $\mathbf{K}^2$ (5.47). As long as we restrict ourselves to the subspace with eigenvalue $K = \Omega/2$, we only have one collective fermion pair operator (9.24)

$$b^+ = \frac{1}{\sqrt{\Omega}} K_+ = \frac{1}{\sqrt{\Omega}} \sum_m \alpha^+_m \beta^+_m, \tag{9.35}$$

with *ph* amplitudes $C_{mm'} = (1/\sqrt{\Omega})\delta_{mm'}$ ($m > 0$). The operator (9.25) is closely connected to K_0

$$a = \frac{1}{\Omega} \sum_m (\alpha^+_m \alpha_m + \beta^+_m \beta_m) = 1 + \frac{1}{\Omega} 2K_0. \tag{9.36}$$

The number of states is $M = 2\Omega$ and the quantities Γ (9.28a, b) are given by

$$\Gamma^{(2)} = +\frac{2}{\Omega}; \qquad \Gamma^{(3)} = -\frac{2}{\Omega^2}; \qquad \Gamma^{(4)} = +\frac{2}{\Omega^3}; \ldots . \tag{9.37}$$

The algebra (9.27) now corresponds to the angular momentum algebra.

The Hamilton operator of the model is given by Eq. (5.45). Expressed in terms of the operators b and a it takes the form

$$H = -\frac{\epsilon}{2}\Omega + \frac{\epsilon}{2}\Omega a - \frac{V}{2}\Omega(b^+ b^+ + bb). \tag{9.38}$$

From the consideration of Section 9.2.2 we could have immediately written down a boson expansion for the operators a and b; however, first we want to study the general case further.

According to the prescription of Beliaev and Zelevinski in the *general case* [BZ 62], we have to start with a set of pure boson operators

$$\left[B_\mu, B_{\mu'}^+\right] = \delta_{\mu\mu'}, \qquad \left[B_\mu, B_{\mu'}\right] = 0, \tag{9.39}$$

and to expand the boson images of the operators b_μ^+ and $a_{\mu\nu}$ in the following way.*

$$(b_\mu^+)_B = \sum_\rho x_{\mu\rho}^{(1)} B_\rho^+ + \sum_{\rho\nu\lambda} x_{\mu\nu\rho\lambda}^{(3)} B_\nu^+ B_\rho^+ B_\lambda + \cdots, \tag{9.40a}$$

$$(a_{\mu\nu})_B = y^{(0)} + \sum_{\rho\lambda} y_{\mu\nu\rho\lambda}^{(2)} B_\rho^+ B_\lambda + \cdots. \tag{9.40b}$$

The coefficients $x^{(i)}$ and $y^{(i)}$ are determined by the condition that the commutation relations (9.27) are satisfied.

Two methods have been developed in the literature to achieve this requirement:

9.2.3.1. The Perturbative Boson Expansion. This method is based on a "*small parameter*" and the commutator algebra is fulfilled order by order. This method was originally proposed by Belyaev and Zelevinskii [BZ 62] and later its properties were much investigated by Marshalek [Ma 71, MH 72, Ma 74]. In these papers the authors have concentrated themselves on fermion pair operators coupled to good angular momentum. The "small parameter" is then

$$\epsilon = \frac{1}{2\bar{j}+1}, \tag{9.41}$$

where $\bar{j}$ is an averaged single-particle angular momentum of the levels in the vicinity of the fermi surface. In fact, this parameter is not very small and the method works much better if one treats collective fermion pairs as in Eq. (9.24). As we shall see, the "small parameter" is in this case the quantity $\Gamma_{\rho\sigma}^{\mu\nu}$ of Eq. (9.28a). If the bosons, which one uses for the expansion, are really collective, that is, if there are many small values of the coefficients C_{kl}^{μ}, the value of $\Gamma^{(2)}$ becomes very small. An example occurs in the Lipkin model [see Eq. (9.37)], where we have $\Gamma^{(2)} \propto \epsilon = 1/\Omega$. In such cases, the higher vertices $\Gamma^{(3)}$, $\Gamma^{(4)}$ of Eq. (9.28b) are of the order $\epsilon^2, \epsilon^3, \ldots$.

The commutator algebra (9.27) is now fulfilled in two steps. First, we can show by a straightforward application of the relations (9.29) or (9.30) that the subalgebra (9.27e) can already be satisifed with a finite expansion

*The following ansatz is certainly not unique, but it is reasonable to assume that the application of a fermion pair operator b_μ^+ increases the number of bosons by just one and that the application of the operator $a_{\mu\nu}$ does not change the boson number.

of Eq. (9.40b):

$$(a_{\mu\nu})_B = \sum_{\rho\sigma} \Gamma^{\mu\rho}_{\nu\sigma} B_\rho^+ B_\sigma, \tag{9.42}$$

that is, $y^{(2)}_{\mu\nu\rho\lambda} = \Gamma^{\mu\rho}_{\nu\lambda}$ and all other coefficients $y^{(n)}$ vanish. In the next step we try to satisfy Eq. (9.27b) at each order in Γ. The zero'th order gives $x^{(1)}_{\mu\nu} = \delta_{\mu\nu}$, the first order is fulfilled by $x^{(3)}_{\mu\nu\rho\sigma} = -\frac{1}{4}\Gamma^{\mu\rho}_{\nu\sigma}$, and so on. After a rather lengthy calculation, which we do not want to present here, because the result can be obtained in a much simpler way by the methods discussed in Section 9.2.6, we end up with the infinite expansion

$$(b_\mu^+)_B = B_\mu^+ - \frac{1}{4}\sum_{\rho\sigma\nu} \Gamma^{\nu\rho}_{\mu\sigma} B_\nu^+ B_\rho^+ B_\sigma + \frac{1}{16} \sum_{\substack{\nu\rho_1\sigma_1 \\ \rho_2\sigma_2}} \Gamma^{\nu\rho_2\rho_1}_{\mu\sigma_1\sigma_2} B_\nu^+ B_{\rho_1}^+ B_{\sigma_1} B_{\rho_2}^+ B_{\sigma_2} + \cdots, \tag{9.43}$$

which is of the form (9.40a) with

$$x^{(2n+1)}_{\mu\nu\rho_1\sigma_1\rho_2\sigma_2\cdots\rho_n\sigma_n} = -\frac{1}{2}\binom{1/2}{n} \cdot \Gamma^{\nu\rho_n\rho_1\rho_{n-1}\rho_2\cdots}_{\mu\sigma_1\sigma_n\sigma_2\sigma_{n-1}\cdots}. \tag{9.44}$$

We can also verify that the expression obeys the relations (9.27c and d) in each order. The boson expansion obtained in this way is not in normal order with respect to the boson vacuum.

In deriving these expressions, at many places we use the completeness relations (9.26) for the coefficients C^μ_{kl} and similar equations of the type (9.30). This means that, in principle, we have to include all indices μ with corresponding solutions of the TDA equation (8.10). Among them there are many non-collective bosons for which the higher terms $\Gamma^{(n)}$ do not drop off rapidly. The whole expansion is therefore only meaningful if we restrict ourselves to a few collective bosons (the subspace spanned by them is called the *collective subspace*), and also if the coefficients Γ have the property that they do not mix these collective bosons very much with the rest of the Hilbert space. For instance, this is the case in the Lipkin model [see Eq. (9.37)], where $SU(2)$ symmetry provides a complete decoupling of the collective subspace. In the general case, it is not always possible to find such a collective subspace.

On the other hand the condition that Γ provides no coupling of the collective subspace and the rest of the Hilbert space is only a *kinematic condition*, that is, it depends only on the property of the underlying boson basis. It is not sufficient to guarantee a meaningful boson expansion. In the introduction we saw that the restriction to a collective subspace is meaningful only in cases where the Hamiltonian provides no serious coupling to the rest of the Hilbert space, that is, in cases where the collective subspace contains to a good approximation some eigenstates of the system. Since this condition depends on the Hamiltonian, it is called a *dynamic condition*.

Both conditions are sometimes fulfilled for group theoretical reasons. Again, an example is the Lipkin model, where we can restrict ourselves to

the collective subspace with quasi-spin $\Omega/2$. In this case we only have one boson B and get from Eq. (9.35ff) and (9.42ff)

$$\begin{aligned} a &= \frac{2}{\Omega} B^+ B, \\ b^+ &= B^+ - \frac{1}{2\Omega} B^+ B^+ B - \frac{1}{8\Omega^2} B^+ B^+ B B^+ B - \cdots, \end{aligned} \tag{9.45}$$

which can be summed up to

$$b^+ = B^+ \sqrt{1 - \frac{1}{\Omega} B^+ B} \ . \tag{9.46}$$

This corresponds exactly to the Holstein–Primakoff representation (9.4) of the angular momentum operators $K_\pm, K_0$. The perturbative boson expansion of Belyaev and Zelevinskii is therefore often called a Holstein–Primakoff expansion. If we expand the Hamiltonian (9.38) up to order $1/\Omega$ we get, after normal ordering,

$$H_B = -\frac{\epsilon}{2}\Omega + \epsilon B^+ B - V\frac{\Omega}{2}\left(\left(1 - \frac{1}{2\Omega}\right) B^+ B^+ - \frac{1}{\Omega} B^+ B^+ B^+ B + \text{h.c.}\right). \tag{9.47}$$

The perturbative boson expansion has the advantage that in lowest order it corresponds exactly to the *RPA approximation* of Section 8.4. To see this, we use non-collective boson operators and stop after the first term in Eq. (9.43), which means that

$$(\alpha_k^+ \alpha_l^+)_B = B_{kl}^+, \qquad (\alpha_k^+ \alpha_l)_B = \sum_p B_{kp}^+ B_{lp}.$$

Inserting these expressions into the Hamiltonian (E.18) and taking into account only terms up to B^+B, B^+B^+, and BB, we find the RPA Hamiltonian of Eq. (8.90) in the quasi-particle representation (8.200):

$$\begin{aligned} H = E_{\text{HF}} &+ \sum_{\substack{k<l \\ k'<l'}} \left\{(E_k + E_l)\delta_{kk'}\delta_{ll'} + H^{22}_{klk'l'}\right\} B_{kl}^+ B_{k'l'} \\ &+ 4 \sum_{\substack{k<l \\ k'<l'}} \left\{H^{40}_{klk'l'} B_{kl}^+ B_{k'l'}^+ + \text{h.c.}\right\} \end{aligned} \tag{9.48}$$

(see, however, footnote on p. 344).

We can now also understand, in a very simple way, why the RPA approach preserves the symmetries [Ma 74]. Since the boson expansion (9.43) is constructed so as to satisfy the commutation relations in each order separately, we find for two arbitrary operators A, B with

$$[A, B] = C, \tag{9.49}$$

$$[A^{(n)}, B^{(1)}] + [A^{(n-1)}, B^{(2)}] + \cdots + [A^{(1)}, B^{(n)}] = C^{(n-1)}, \tag{9.50}$$

where $A^{(n)}$, $B^{(n)}$, and $C^{(n)}$ are the nth order terms of the boson parts of the operators A, B, C, which contain n operators B or B^+. Since the order

is determined by the expansion parameter ϵ, these operators are not in normal order. Reordering them would imply that terms with a fixed number of operators B or B^+ contain contributions in a different order in ϵ.

For a symmetry operator P which commutes with H, we find from Eq. (9.50) in the case $n=2$*

$$[H^{(2)}, P^{(1)}]=0, \tag{9.51}$$

which is exactly the RPA equation (8.104) for the zero frequency mode $P^{(1)}$. These arguments can easily be extended to higher order in the boson expansion. The practical applicability of this advantage, however, is restricted by the fact that it is only true if the commutator algebra is fulfilled exactly. In higher orders we usually need additional drastic approximations —for instance, the restriction to only a few collective bosons—which destroy this property.

Mathematically, the RPA Hamiltonian (9.48) has the same structure for bosons as the general single-particle Hamiltonian of Eq. (7.59) for fermions. Both are diagonalized by Bogoliubov transformations [Eqs. (7.1) and (8.91); see also Appendix E.5] resulting in the HFB equations (7.42) and the RPA equations (8.70). The only difference is that the HFB equations are nonlinear. They contain a part of the fourth-order term in the fermion Hamiltonian (i.e., of the two-body interaction). Going to higher orders in the boson expansion in Eq. (9.48), we can proceed as in Eqs. (7.57f.) and bring all terms that we want to take into account into normal order with respect to the new boson vacuum. All quadratic terms B^+B, B^+B^+, and BB can be brought into the form (8.90) with single-boson fields depending on the boson densities. We thus end up with a *self-consistent random phase approximation* (SCRPA) [SE 73, Ma 76a].

* The term $H^{(1)}$ corresponds to H^{20} in Eq. (7.36) and is assumed to vanish.

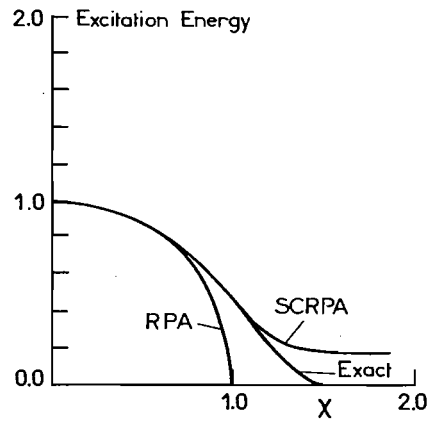


Figure 9.2. The self-consistent RPA is compared to the usual RPA and the exact solution of the Lipkin model as a function of the coupling strength χ. (From [SE 73].)

Figure 9.2 shows its solution in the case of the Lipkin model (see Sec. 5.4). We see clearly how the breakdown of RPA in the region of the phase transition (as discussed in Sec. 8.8.2) is avoided by the treatment of higher terms in the boson expansion. For very large particle numbers, SCRPA and RPA begin to coincide, indicating that in very large systems there can be sharp phase transitions, whereas in finite systems the phase transition is always smeared out to a certain extent (see Chap. 11).

9.2.3.2. Restriction to a Few Collective Bosons. In some cases there are physical reasons for restrictions to a few collective bosons, such as quadrupole oscillations and pairing vibrations. The expansion (9.43) does not then satisfy the commutator algebra exactly. Following Sørensen [Sø 67], we can nevertheless use the ansatz (9.40) up to a fixed number of boson operators with variable parameters $x^{(1)}$, $x^{(3)},\ldots x^{(2n+1)}$ and $y^{(0)}$, $y^{(1)},\ldots y^{(2n)}$ as unknowns. By inserting this ansatz into the algebra, evaluating the commutators and then equating the coefficients of the same types of operators, we get a set of equations (which are generally nonlinear) for the coefficients $x^{(i)}$ and $y^{(i)}$. For example, in the case of only one boson,

$$\begin{aligned} 1-y^{(0)} &= |x^{(1)}|^2, \\ -y^{(2)} &= 2(|x^{(3)}|^2 + x_1^* x_3 + x_3^* x_1), \\ &\vdots \qquad\qquad \vdots \end{aligned} \tag{9.52}$$

The evaluation of these equations is possible, and has been carried out by Sørensen [Sø 67, 68a+b, 69, 70, 71, 73] to fourth order and also by Kishimoto and Tamura [KT 72, 76] to sixth order in the Hamiltonian (see also [SP 77b]). However, the parameter $y^{(0)}$ is always open. A change of this parameter corresponds to a unitary transformation within the boson space [Sø 68b]. It has an influence on the convergence properties and has been used in different ways.

This kind of boson expansion is not a perturbative expansion, in the sense that the coefficients in a certain order are not changed if further terms are included. In this sense it may be better adapted to the collective bosons under consideration. The dynamical condition that the Hamiltonian should not couple the collective subspace with the rest of the Hilbert space is not touched by these considerations.

9.2.4 The Boson Expansion of Marumori

Marumori et al. [MYT 64a+b] propose a method which starts with a one-to-one mapping of states in the fermion space to states in the physical subspace of the boson space. The procedure is carried out for non-collective bosons, and we shall give a short discussion of the mathematical details in Section 9.2.6. Since, in this case, the convergence properties are very poor, Holzwarth et al. [LH 75] have used, in practical applications, a somewhat modified version using a few collective bosons. The same method has also been applied to exactly soluable models [PKD 68, Kl 69]. Starting from collective fermion pair operators b_μ^+ of the form (9.24), we

introduce the normalized $2n$-fermion states

$$|n\rangle = \mathcal{N}_F(n)|\,\mu_1 \ldots \mu_n\rangle = \mathcal{N}_F(n) b^+_{\mu_1} \ldots b^+_{\mu_n}|0\rangle, \tag{9.53}$$

where $\mathcal{N}_F(n)$ is a normalization constant. These states are, in general, not orthogonal. We will assume, however, that they are linearly independent, which is certainly true if only a few collective operators are taken into account and the number n is not too large (i.e., the whole number must not exceed the corresponding fermion space). The norm matrix $N_{nn'}$ is given by

$$N_{nn'} = \langle \mu_1 \ldots \mu_n|\,\mu'_1 \ldots \mu'_n\rangle. \tag{9.54}$$

It is closely related to the quantities Γ of Eq. (9.28).

$$\begin{aligned} N_{11}: &\quad \langle \mu|\,\mu'\rangle = \delta_{\mu\mu'}, \\ N_{22}: &\quad \langle \mu\nu|\,\mu'\nu'\rangle = \delta_{\mu\mu'}\delta_{\nu\nu'} + (\mu \leftrightarrow \nu) - \Gamma^{\mu\nu}_{\mu'\nu'}, \\ N_{33}: &\quad \langle \mu\nu\rho|\,\mu'\nu'\rho'\rangle = \delta_{\mu\mu'}\delta_{\nu\nu'}\delta_{\rho\rho'} - \delta_{\rho\rho'}\Gamma^{\mu\nu}_{\mu'\nu'} - \Gamma^{\mu\nu\rho}_{\mu'\nu'\rho'} \\ &\qquad\qquad (+\text{several permutations}). \end{aligned} \tag{9.55}$$

The diagonal matrix elements define the norm factors; for instance, we get for the *two phonon norm*

$$\mathcal{N}_F(2) = (1 + \delta_{\mu\nu} - \Gamma^{\mu\nu}_{\mu\nu})^{-1/2}. \tag{9.56}$$

Marumori now introduces a boson space spanned by the normalized orthogonal states

$$|n) = \mathcal{N}_B(n)|\,\mu_1 \ldots \mu_n) = \mathcal{N}_B(n) B^+_{\mu_1} \ldots B^+_{\mu_n}|0). \tag{9.57}$$

$\mathcal{N}_B$ is a trivial factor of the form $(n_1!n_2!\ldots)^{-1/2}$ (see C.19). The unitary operator

$$U = \sum_{nn'} |n) N^{-1/2}_{nn'} \langle n'| \cdot \mathcal{N}_F^{-1}(n') \tag{9.58}$$

maps the space of fermions into this space.

In many practical applications (see Sec. 9.2.8), the matrix N is diagonal, otherwise the application of $N^{-1/2}$ means an orthogonalization. We therefore restrict ourselves to this case and gain, for the transformation of operators to the boson space,

$$H_B = UHU^+ = \sum_{nn'} \langle n|H|n'\rangle |n)(n'|. \tag{9.59}$$

The operators $|n)(n'|$ in the boson space can be expressed by the projector onto the boson vacuum*

$$|0)(0| =: \exp\Big(-\sum_\mu B^+_\mu B_\mu\Big) := 1_B - \sum_\mu B^+_\mu B_\mu + \frac{1}{2!}\sum_{\mu\nu} B^+_\mu B^+_\nu B_\mu B_\nu \ldots \tag{9.60}$$

*The dots : mean normal ordering of the operators. The relation (9.60) is easy to prove in the case of one boson:

$$B|0)(0| = B(1 - B^+B + \tfrac{1}{2}B^+B^+BB - \cdots) = B - B + B^+BB - B^+BB + \cdots = 0.$$

in an obvious shorthand notation

$$|n)(n'| = \mathfrak{N}_B(n)\mathfrak{N}_B(n')(B^+)^n(1 - \ldots B^+B + \ldots B^+B^+BB - \cdots)(B^+)^{n'}. \tag{9.61}$$

If $|n)$ represent only a few collective states, the Hamiltonian (9.59) corresponds only to the collective part of the many-body Hamiltonian, and its eigenstates have a physical meaning only in cases where the full Hamiltonian does not couple the collective subspace to the rest of the Hilbert space. Therefore, the choice of proper bosons requires physical intuition.

In Eq. (9.59), the collective Hamiltonian is expressed as a normal ordered power series of boson operators. It contains the fermion matrix elements of the Hamiltonian in the collective subspace and the expansion of the boson-vacuum projector (9.60). As we shall see, it is usually a good approximation to take into account, in the sum (9.59), only low values of n and n': $n + n' \leqslant n_{\max}$. Therefore, for practical application we need only low-order fermion matrix elements $\langle n|H|n'\rangle$. This is actually the *main practical advantage* of the boson theories (see below). They are calculated in the basis of collective fermion pairs b_μ^+ using the representation (9.32) of the Hamiltonian. Considering, for example, the matrix element $\langle n+2|H|n\rangle$, we find

$$\langle \mu_1 \ldots \mu_{n+2}|H|\nu_1 \ldots \nu_n\rangle = \sum_{\rho\sigma} V^{40}_{\rho\sigma}\langle \mu_1 \ldots \mu_{n+2}|\rho\sigma\nu_1 \ldots \nu_n\rangle. \tag{9.62}$$

With the assumption that the Hamiltonian does not scatter out of the collective space, the sum over ρ and σ is restricted to the few collective indices only, and we are left with the problem of calculating the norm matrix elements (9.54), which in practical applications is a rather tedious problem, because they take into account the corrections of the Pauli principle in the form of the quantities Γ in Eqs. (9.28).

At this step one often uses the additional approximation that the commutation relation (9.27c) is already fulfilled within the collective space, if the sum over ρ on the r.h.s. only runs over states in this space. It means that the double commutator

$$\left[b_\mu, \left[b_{\mu'}, b^+_{\nu'}\right]\right] = -\sum_\rho \Gamma^{\mu\mu'}_{\nu'\rho} b_\rho \cdot \tag{9.63}$$

does not scatter out of the collective space, an approximation which we have already used in the framework of the Belyaev–Zelevinskii approximation.

Using Eqs. (9.27b, c) we then can derive a *recursion relation* (see also [HJJ 76, IST 77, SP 78]):

$$\begin{aligned}\langle \mu_1 \ldots \mu_n|\nu_1 \ldots \nu_n\rangle &= \delta_{\mu_1\nu_1}\langle \mu_2 \ldots \mu_n|\nu_2 \ldots \nu_n\rangle \\ &\quad + \sum(\mu_1 \text{ exchanged with all } \mu_2 \ldots \mu_n) \\ &\quad - \sum_\lambda \Gamma^{\mu_1\mu_2}_{\nu_1\lambda}\langle \lambda\mu_3 \ldots \mu_n|\nu_2 \ldots \nu_n\rangle \\ &\quad - \sum[\mu_1\mu_2 \text{ exchanged with all pairs in } (\mu_1 \ldots \mu_n)].\end{aligned} \tag{9.64}$$

Starting from the two-phonon norm (9.55) we are therefore able to express all higher norm matrix elements by $\Gamma^{(2)}$. Holzwarth et al. [HJJ 76] have checked the validity of this approximation in realistic transitional nuclei and found very good agreement with exact expressions for the norm matrix elements.

We will see in Section 9.2.6 that the quantities $\langle \mu_1 \dots \mu_n | \nu_1 \dots \nu_n \rangle$ are proportional to the matrix elements of the projection operator $\hat{P}$ onto the physical subspace (see Eq. 9.99). They take into account the fact that the collective two-quasi-particle operators b_μ^+ are not real bosons. The fermion spaces spanned by these operators are finite, whereas the boson space spanned by the corresponding boson operators is infinite. Therefore, we find that the quotient of fermion to boson norm,

$$\nu(n) = \frac{\langle \mu_1 \dots \mu_n | \mu_1 \dots \mu_n \rangle}{(\mu_1 \dots \mu_n | \mu_1 \dots \mu_n)}, \tag{9.65}$$

becomes very small for large n-values. This means that the effective fermion space is truncated drastically by the Pauli principle. In particular, we find that the matrix elements $(n+2|H|n)$, which couple states with different boson numbers, become small for large n-values. As a consequence, we can restrict the sum in Eq. (9.59) to low values of n and n'. In many cases [HJJ 76, SP 77c], a Marumori expansion of the Hamiltonian up to fourth or sixth order is therefore a very good approximation.

Summarizing these results, we see two conditions under which the collective Marumori expansion is valid:

(i) The first condition is a purely *kinematic* one (i.e., it has only to do with the collective operators b_μ^+ and the quantities Γ): The fermion norm matrix elements have to drop dramatically for large values of n, with the consequence that higher-order terms are cut down by the Pauli principle

(ii) The second condition is a *dynamic* one (i.e., it has to do with the properties of the Hamiltonian H): The collective subspace spanned by the vectors $|\mu_1 \dots \mu_n\rangle$ under consideration has to coincide with an approximate eigenspace of the many-body Hamiltonian, that is, H should couple only weakly to the rest of the Hilbert space.

The first condition seems to be in contradiction to the condition of very collective bosons and small Γ-values in the case of perturbative boson expansions (see Sec. 9.2.3). For small Γ-values, the effective fermion space is indeed rather large. In fact, the first condition would be best fulfilled for a few pure two-quasi-particle operators, where the effective fermion space is very small. In this case, however, the second condition is hardly fulfilled. Only for *collective* bosons is the corresponding fermion space large enough that we can find cases where the exact eigenstates of the Hamiltonian can be represented in this way. It turns out, however, that the effects of the Pauli principle are still strong enough in these cases to guarantee rapid convergence (see [HJJ 76]).

As an example, we again investigate the *Lipkin model* of Section 5.4 (for

details, see [PKD 68]). From Eqs. (9.37) and (9.64), we gain the recursion relation for the norm,

$$\langle (b)^n (b^+)^n \rangle = \langle (b)^{n-1} (b^+)^{n-1} \rangle \left(n - \frac{n}{2}(n-1)\frac{2}{\Omega} \right), \tag{9.66}$$

which in this case is exact. For the normalization factors it yields

$$\mathfrak{N}_F(n) = \sqrt{\frac{(\Omega - n)!}{\Omega! n!} \Omega^n}\ , \qquad \mathfrak{N}_B = \frac{1}{\sqrt{n!}}\ , \tag{9.67}$$

and for the Hamilton (9.38) up to four boson terms we have, from Eqs. (9.59f),

$$H_B = -\frac{\epsilon}{2}\Omega + \epsilon B^+ B - V\frac{\Omega}{2}\left[\sqrt{1 - \frac{1}{\Omega}}\ B^+ B^+ \right.$$

$$\left. - \left(\sqrt{1 - \frac{1}{\Omega}} - \sqrt{\left(1 - \frac{1}{\Omega}\right)\left(1 - \frac{2}{\Omega}\right)} \right) B^+ B^+ B^+ B + \text{h.c.} \right]. \tag{9.68}$$

We see that the Marumori representation does not correspond to an expansion in $1/\Omega$. Each coefficient contains all orders in $1/\Omega$. If we expand them to a certain order in $1/\Omega$, we get the equivalent Belyaev–Zelevinskii terms of Eq. (9.47). On the other hand, we get the Marumori Hamiltonian by taking all orders of the Belyaev–Zelevinskii expansion, bringing them into normal order, and summing up the coefficients belonging to a fixed number of boson operators, because both expansions are equivalent in infinite order (see Sec. 9.2.6). The advantage of the Marumori expansion is that it takes the Pauli principle into account exactly in each order n.

In fact, the diagonalization of the Marumori operator in a boson space with a maximal number of $N_{\max}$ bosons corresponds to the diagonalization of the fermion Hamiltonian in the space spanned by the fermion states $|n\rangle$, with $n \leqslant N_{\max}$. In the latter case, we would have to calculate the matrix elements $\langle n|H|n'\rangle$. From Eq. (9.59), we see that they are obtained as matrix elements $(n|H_B|n')$ of the Marumori operators in a boson space. The advantage of the transcription to a boson picture is the fact that the sum over n can be restricted to a small value $n_{\max}$ under the conditions discussed above. This means that the higher order matrix elements $(n|H_B|n')$ are expressed by sums containing only the low-order fermion matrix elements.

If we include all fermion pairs b_μ^+ (also non-collective ones) the Marumori method corresponds exactly to a shell model calculation in an $2N_{\max}$-quasi-particle space. In lowest order, therefore, we get not the RPA, but the TDA approximation.

9.2.5 The Boson Expansion of Dyson

The boson representations discussed so far are infinite expansions which converge only under certain conditions, such as a high collectivity of the underlying fermion pairs. In some cases—in particular, near closed shells —we must also take into account other types of less collective two-fermion pairs and their mixing with the collective subspace. In such cases, we must allow for several types of bosons with a different degree of collectivity. Nevertheless we hope that, at least for low-lying states, convergence can be found in the number m of different bosons.

This is quite a different type of convergence [(vertical direction in Fig. 9.3)]. It can be investigated by the boson representation of Dyson, because this method ends up with a finite boson Hamiltonian. Figure 9.3 also shows the cases that have been under investigation up to the present. Horizontal convergence with a few collective bosons [HJJ 76] in the transition region of soft nuclei has been tested, as has vertical convergence for many different Bosons—but only a few of them at one time have been tested in the region of the magic nucleus ^{208}Pb [IRS 80].

In general, there will most probably be cases in which many different types of bosons, as well as large numbers of them, are also needed, a region (characterized by a question mark in Fig. 9.3) which is difficult to investigate.

The Dyson representation is based on the fact that the commutator algebra (9.27) can be satisfied by the mapping* [RS 77a] .

$$b_\mu^+ \to \tilde{B}_\mu^+ = B_\mu^+ - \frac{1}{2}\sum_{\nu\rho\sigma} \Gamma_{\mu\sigma}^{\nu\rho} B_\nu^+ B_\rho^+ B_\sigma, \tag{9.69a}$$

$$b_\mu \to B_\mu, \tag{9.69b}$$

$$a_{\mu\nu} \to \sum_{\rho\sigma} \Gamma_{\nu\sigma}^{\mu\rho} B_\rho^+ B_\sigma. \tag{9.69c}$$

It is a generalization of the Dyson representation (9.9) of the angular momentum algebra. By mapping the fermion vacuum $|0\rangle$ onto the boson vacuum $|0)$, we obtain the corresponding representation of the vectors in the Hilbert space.

From the finite expressions (9.32) and (9.69) we get a finite boson Hamiltonian, which contains terms up to sixth order in the operators B and B^+. The operators H^{11} and H^{22} conserve the boson number. If we assume that the coefficients C_{kl}^{μ} of Eq. (9.24) are determined by the TDA equation (8.10), we get after a straightforward calculation,

$$V_{\mu\nu}^{22} = \delta_{\mu\nu}\Omega_\mu - \sum_k \Gamma_{\nu k}^{\mu k} \cdot E_k \tag{9.70}$$

and

$$H^{11} + H^{22} = \sum_\mu \Omega_\mu B_\mu^+ B_\mu + \frac{1}{4}\sum_{\mu\nu\rho\sigma} W_{\mu\nu\rho\sigma} B_\mu^+ B_\nu^+ B_\rho B_\sigma, \tag{9.71}$$

* For an early version of this representation, see also [Am 67].

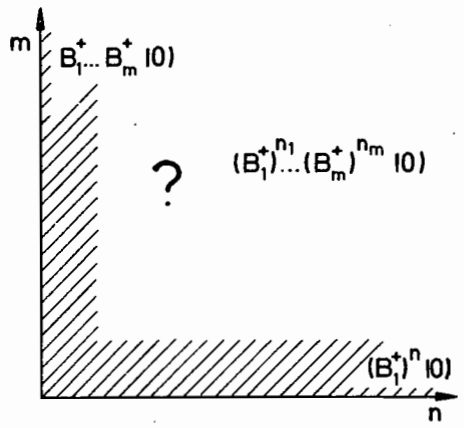


Figure 9.3. Schematic representation of the two kinds of convergence we are faced with in boson expansions. m gives the number of different boson types $B_1^+ \ldots B_m^+$ and n gives the number of bosons of one type $(B_1^+)^n$.

with the matrix elements

$$W_{\mu\nu\rho\sigma} = \sum_{klpq} C_{kl}^{\mu *} C_{pq}^{\nu *} C_{kp}^{\rho} C_{lq}^{\sigma} (E_k + E_l + E_p + E_q - \Omega_\rho - \Omega_\sigma). \qquad (9.72)$$

The other terms of the Hamiltonian violate the boson number and can be found in [RS 77a].

A disadvantage of the Dyson-representation is the fact that it violates Hermiticity, as we see, for instance, from the operator W in Eq. (9.71). The images of the bra and ket vectors

$$|\mu_1 \ldots \mu_n\rangle \rightarrow \tilde{B}_{\mu_1}^+ \ldots \tilde{B}_{\mu_n}^+ |0), \qquad \langle \mu_1 \ldots \mu_n| \rightarrow (0| B_{\mu_n} \ldots B_{\mu_1}$$

no longer form an orthogonal set.

In analogy to (9.54), we get for the norm matrix N:

$$N(\mu_1 \ldots \mu_n, \mu_1' \ldots \mu_n') = (0| B_{\mu_1} \ldots B_{\mu_n} \tilde{B}_{\mu_1'}^+ \ldots \tilde{B}_{\mu_n'}^+ |0). \qquad (9.73)$$

Since there are fast computer programs for the diagonalization of non-Hermitian matrices, this is not an essential shortcoming [RS 74b, SWR 76, Sch 76]. We will discuss some of the mathematical details in Section 9.2.6 and 9.3.6.

The method is extremely simple if one can neglect the coupling of subspaces with different boson numbers. In such cases the Hamiltonian is represented by $H^{11} + H^{22}$. It can be expressed entirely by the TDA amplitudes C_{kl}^{μ}, the TDA energies Ω_μ, and the quasi-particle energies E_k.

9.2.6 The Mathematical Background

In this section we want to discuss some mathematical properties of the boson representations treated so far. We can thus clarify the mathematical connections between the different methods. The reader who is not interested in such details may skip this section, which is highly mathematical. On the other hand, we are not

able to present all steps in the calculations, which are often rather lengthy, but refer the reader to the corresponding literature.

To understand the mathematical structure of the boson representations it is useful to work with "non-collective" bosons $B_{kl} = -B_{lk}$. Their index is the index of a fermion pair (9.22) and they obey the boson commutation relations

$$[B_{kl}, B_{k'l'}] = 0; \qquad [B_{kl}, B^+_{k'l'}] = \delta_{kk'}\delta_{ll'} - (k \leftrightarrow l). \tag{9.74}$$

Starting from a complete orthogonal basis in the many-fermion space

$$|n\rangle = |k_1 l_1 \ldots k_n l_n\rangle = \alpha^+_{k_1}\alpha^+_{l_1} \ldots \alpha^+_{k_n}\alpha^+_{l_n}|0\rangle, \tag{9.75}$$

we can define a corresponding orthogonal set of states in the boson space [MYT 64a]

$$|n) = \frac{1}{\sqrt{(2n-1)!!}} \sum_P (-)^P P B^+_{k_1 l_1} \ldots B^+_{k_n l_n}|0), \tag{9.76}$$

where P runs over all permutations of the indices $k_1 \ldots l_n$, which provide new combinations of bosons $B^+_{k_1 l_1} \ldots B^+_{k_n l_n}$ (i.e., $B^+_{12}B^+_{34}$ and $B^+_{34}B^+_{21}$ are each counted only once); there are $(2n)!/(n!2^n) = (2n-1)!!$ terms.

The states $|n)$ of Eq. (9.76) span a linear subspace of the boson space, the so-called *physical subspace*. As we shall see, there is a one-to-one correspondence between this subspace and the fermion space. The rest of the boson space contains states $B^+_{k_1 l_1} \ldots B^+_{k_n l_n}|0)$, which are not completely antisymmetrized in the indices $k_1 \ldots l_n$. They therefore violate the Pauli principle. States that have components of this type can provide spurious contributions to the expectation values of physical operators.

Janssen et al. [JDF 71] have shown that the basis states in Eq. (9.76) can also be written as

$$|n) = \frac{1}{\sqrt{(2n-1)!!}} \langle 0| \exp\left\{ \frac{1}{2} \sum_{kl} B^+_{kl} \alpha_l \alpha_k \right\} \alpha^+_{k_1}\alpha^+_{l_1} \ldots \alpha^+_{k_n}\alpha^+_{l_n}|0\rangle |0) \tag{9.77}$$

$$= \frac{1}{\sqrt{(2n-1)!!}} \tilde{B}^+_{k_1 l_1} \ldots \tilde{B}^+_{k_n l_n}|0), \tag{9.78}$$

with the non-collective Dyson operators (9.69a)

$$\tilde{B}^+_{kl} = B^+_{kl} - \sum_{pq} B^+_{kp} B^+_{lq} B_{pq}. \tag{9.79}$$

An example is the two-boson state

$$|2) = \frac{1}{\sqrt{3}} \tilde{B}^+_{k_1 l_1} \tilde{B}^+_{k_2 l_2}|0) = \frac{1}{\sqrt{3}} \left(B^+_{k_1 l_1} B^+_{k_2 l_2} - B^+_{k_1 k_2} B^+_{l_1 l_2} + B^+_{l_1 k_2} B^+_{k_1 l_2} \right)|0). \tag{9.80}$$

The projector $\hat{P}$ onto the physical subspace has the form

$$\hat{P} = \sum_n |n)(n| = \sum_{n=0}^{\infty} \frac{1}{(2n)!} \frac{1}{(2n-1)!!} \sum_{\substack{k_1 \ldots k_n \\ l_1 \ldots l_n}} \tilde{B}^+_{k_1 l_1} \ldots \tilde{B}^+_{k_n l_n}|0)(0|\tilde{B}_{k_n l_n} \ldots \tilde{B}_{k_1 l_1}. \tag{9.81}$$

After a straightforward application of the relations

$$\begin{aligned} &\sum_{\substack{k_1 \ldots k_n \\ l_1 \ldots l_n}} \tilde{B}^+_{k_1 l_1} \ldots \tilde{B}^+_{k_n l_n}|0)(0|\tilde{B}_{k_n l_n} \ldots \tilde{B}_{k_1 l_1} \\ &\quad = (2n-1)!! \sum_{\substack{k_1 \ldots k_n \\ l_1 \ldots l_n}} \tilde{B}^+_{k_1 l_1} \ldots \tilde{B}^+_{k_n l_n}|0)(0|B_{k_n l_n} \ldots B_{k_1 l_1} \end{aligned} \tag{9.82}$$

and

$$(0|\tilde{B}_{p_n q_n}\dots\tilde{B}_{p_1 q_1}\tilde{B}^{+}_{k_1 l_1}\dots\tilde{B}^{+}_{k_n l_n}|0)=(2n-1)!!(0|B_{p_n q_n}\dots B_{p_1 q_1}\tilde{B}^{+}_{k_1 l_1}\dots\tilde{B}^{+}_{k_n l_n}|0), \quad (9.83)$$

we obtain for the quantities

$$(B^{+}B)_{kl}=\sum_{q} B^{+}_{ql}B_{qk} \quad \text{and} \quad \hat{N}=\frac{1}{2}\sum_{kl} B_{kl}B_{kl} \quad (9.84)$$

the following properties

$$[\hat{P},\tilde{B}^{+}_{kl}]=0, \qquad \hat{P}B_{kl}\hat{P}=B_{kl}\hat{P}, \qquad [\hat{P},(B^{+}B)_{kl}]=0, \quad (9.85)$$

$$\tilde{B}^{+}_{kl}\hat{P}=\hat{P}B^{+}_{kl}(1+2\hat{N}), \qquad 2\hat{P}B^{+}_{kl}\hat{N}=-\hat{P}(B^{+}\cdot(B^{+}B))_{kl}. \quad (9.86)$$

There are several possible ways of *mapping* the fermion space onto the physical subspace of the bosons. Usui [Us 60] used the operator

$$\overline{U}=\langle 0|\exp\left\{\frac{1}{2}\sum_{kl} B^{+}_{kl}\alpha_l\alpha_k\right\}|0). \quad (9.87)$$

Inserting a complete set (9.75) in the fermion space and using Eq. (9.77), it can be written as

$$\overline{U}=\sum_{n}\sqrt{(2n-1)!!}\;|n)\langle n|. \quad (9.88)$$

It is obviously not unitary. We therefore have to define a conjugate operator

$$\tilde{U}=\sum_{n}\frac{1}{\sqrt{(2n-1)!!}}|n\rangle(n|=\overline{U}^{+}((2\hat{N}-1)!!)^{-1}, \quad (9.89)$$

where $\hat{N}$ is the boson number operator (9.84). If 1_F is the unity operator in the fermion space, we have

$$\tilde{U}\overline{U}=1_F \quad \text{and} \quad \overline{U}\tilde{U}=\hat{P}. \quad (9.90)$$

The fermion pair operators (9.22) are mapped in the following way.

$$\begin{aligned} b^{+}_{kl} &\to \overline{U}\alpha^{+}_{k}\alpha^{+}_{l}\tilde{U}=\tilde{B}^{+}_{kl}\cdot\hat{P}, \\ b_{kl} &\to \overline{U}\alpha_l\alpha_k\tilde{U}=B_{kl}\cdot\hat{P}, \\ a_{kl} &\to \overline{U}\alpha^{+}_{l}\alpha_k\tilde{U}=(B^{+}B)_{kl}\cdot\hat{P}. \end{aligned} \quad (9.91)$$

This transformation obviously corresponds to the *Dyson* representation discussed in Section 9.2.5. The corresponding Hamiltonian has the form

$$\overline{U}H\tilde{U}=H_B\cdot\hat{P} \quad (9.92)$$

with

$$\begin{aligned} H_B=&\sum_{k}E_k(B^{+}B)_{kk}+\sum_{klpq}\left(H^{40}_{klpq}\tilde{B}^{+}_{kl}\tilde{B}^{+}_{pq}+H^{40*}_{klpq}B_{kl}B_{pq}\right) \\ &+\sum_{klpq}\left(H^{31}_{klpq}\tilde{B}^{+}_{kl}(B^{+}B)_{qp}+H^{31*}_{klpq}(B^{+}B)_{pq}B_{kl}\right) \\ &+\frac{1}{4}\sum_{klpq}H^{22}_{klpq}\tilde{B}^{+}_{kl}B_{pq}. \end{aligned} \quad (9.93)$$

As long as we restrict ourselves to physical wave functions we can neglect the projector and solve the Schrödinger equation

$$H_B|\Psi)=E|\Psi). \quad (9.94)$$

However, we have to be aware that there are also spurious solutions $|\Psi_s)$ which come from a violation of the Pauli principle. . Since $\hat{P}$ does not commute with H_B, they can have components in the physical subspace, that is, $\hat{P}|\Psi_s) \neq 0$. To recognize them we have to apply the criterion $(1-\hat{P})|\Psi_s) \neq 0$. In cases where the eigenvalue problem is approximated by restricting the number of collective bosons they can mix with the physical solutions. It is therefore often meaningful to decompose

$$H_B = H_B^0 + V_B, \tag{9.95}$$

with $H_B^0 = \sum_k E_k (B^+ B)_{kk}$. From Eq. (9.85) we see that H_B^0 commutes with $\hat{P}$. The physical solutions of Eq. (9.94) therefore also obey the equation

$$(H_B^0 + \hat{P} V_B)|\Psi) = E|\Psi). \tag{9.96}$$

From a decomposition of any solution $|\Psi) = \hat{P}|\Psi) + (1-\hat{P})|\Psi)$ into a physical and a spurious part, we see that its spurious part is an eigenvector of H_B^0 with the eigenvalue E.

$$H_B^0 (1-\hat{P})|\Psi) = E(1-\hat{P})|\Psi), \tag{9.97}$$

that is, the unphysical solutions of Eq. (9.96) have the eigenvalues of H_B^0. They are given by

$$E_{k_1} + E_{l_1} + E_{k_2} + E_{l_2} + \cdots. \tag{9.98}$$

The lowest spurious eigenfunctions of H^0 are two boson states of the type $B_{12}^+ B_{12}^+ |0)$. The lowest spurious solutions of (9.96) therefore lie at the free four-quasi-particle energies. This is usually much higher than the interesting low-lying part of the spectrum and an approximate solution of Eq. (9.96) does not provide serious mixing between physical and unphysical states.

In practical applications, it is meaningful to start with collective bosons. The form of the Hamiltonian H_B has already been derived in Section 9.2.5. To apply Eq. (9.96) we only need the matrix elements of the projector $\hat{P}$. We see from Eq. (9.81f) that they are given by the norm (9.54) or (9.73).

$$\begin{aligned}(\mu_1, \ldots, \mu_n|\hat{P}|\mu_1' \ldots \mu_n') &= (0|B_{\mu_1} \ldots B_{\mu_n} \hat{P} B_{\mu_1'}^+ \ldots B_{\mu_n'}^+|0) \\ &= \frac{1}{(2n-1)!!} \langle \mu_1 \ldots \mu_n|\mu_1' \ldots \mu_n'\rangle. \end{aligned} \tag{9.99}$$

If we have physical eigensolutions of Eq. (9.94) or (9.96) we can calculate the matrix elements of a transition operator Q by transforming to the boson space

$$\bar{U} Q \tilde{U} = Q_B \cdot \hat{P} \tag{9.100}$$

and get

$$\begin{aligned}\langle \Psi|Q|\Psi\rangle &= \langle \Psi|\tilde{U}\bar{U}Q\tilde{U}\bar{U}|\Psi\rangle \, (\Psi|Q_B \hat{P}|\Psi) \\ &= (\Psi|Q_B|\Psi). \end{aligned} \tag{9.101}$$

This shows that we can calculate all interesting quantities in the boson space without transforming back to the fermion space as long as we know that $|\Psi)$ contains no spurious components. This can be achieved by diagonalizing the norm matrix (9.99) and restricting to the space with non-vanishing eigenvalues.

Marumori avoids non-Hermitian operators by a unitary mapping

$$U = \sum_n |n\rangle(n| = ((2\hat{N}-1)!!)^{-1/2} \bar{U} \tag{9.102}$$

and obtains from Eqs. (9.91) and (9.86),

$$\begin{aligned} b_{kl}^{+} &\to U\alpha_k^{+}\alpha_l^{+}U^{+} = \tilde{B}_{kl}^{+}\,(1+2\hat{N})^{-1/2}\hat{P} = \hat{P}(B^{+}\sqrt{1-B^{+}B}\,)_{kl}\hat{P}, \\ b_{kl} &\to U\alpha_l\alpha_k U^{+} = (1+2\hat{N})^{-1/2}\tilde{B}_{kl}\hat{P} \;= \hat{P}(\sqrt{1-B^{+}B}\cdot B)_{kl}\hat{P}, \\ a_{kl} &\to U\alpha_l^{+}\alpha_k U^{+} = \hat{P}(B^{+}B)_{kl}\hat{P}. \end{aligned} \tag{9.103}$$

In all practical applications, the square roots in Eq. (9.102) have to be replaced by a Taylor series. Therefore, the Marumori theory ends up with an infinite series, even if we restrict ourselves to the physical subspace.

The expansion coefficients of the square root are $(-)^n\binom{1/2}{n}$. Equation (9.103) therefore corresponds exactly to the *perturbative boson expansion* in Eq. (9.43). If we go to infinite order it is equivalent to the method of Marumori. The latter is obtained after normal ordering. Stopping at a finite order gives differences between both these cases [see, e.g., the Lipkin model in Eqs. (9.47) and (9.68)].

We also see that a formal expansion of $\sqrt{1-B^{+}B}$ does not converge in the case of non-collective bosons [Ok 74]. Only by introducing collective bosons and an appropriate decoupling of the corresponding subspace can convergence as discussed in Section 9.2.4 be achieved.

9.2.7 Methods Based on *pp*-Bosons

So far we have discussed several methods based on bosons in the vicinity of the fermi surface. These bosons were, to a good approximation, given by *ph* pairs or two-quasi-particle pairs. As a natural consequence, the boson number was not conserved in all these methods. The corresponding Hamiltonians included terms of the form $B^{+}B^{+}$, $B^{+}B^{+}B$.

We now start from quite a different approach, namely from the bare vacuum $|-\rangle$ (or, in practical cases, also from a completely inert core) and add pairs of two particles coupled to an approximate boson. An example are Cooper pairs in a single j-shell* [Eq. (6.3)]

$$S_{+} \propto \sum_{m} C^{j\;\;j\;\;0}_{m-m\;\;0} a_{jm}^{+} a_{j-m}^{+} \tag{9.104}$$

or quadrupole pairs

$$D_{\mu}^{+} \propto \sum_{mm'} C^{j\;\;j\;\;2}_{mm'\mu} a_{jm}^{+} a_{jm'}^{+}. \tag{9.105}$$

In the seniority scheme (Sec. 6.2), we can represent the ground state as a product of Cooper pairs [Eq. (6.19)]

$$|0\rangle \propto (S_{+})^{N/2}|-\rangle$$

and some excited states by replacing one S_{+} operator by a D_{μ}^{+} [Eq. (6.27)]:

$$|2\mu\rangle \propto (S_{+})^{N/2-1} D_{\mu}^{+}|-\rangle. \tag{9.106}$$

In the general case, we can start with collective two-particle pairs

$$b_{\tau}^{+} = \frac{1}{2}\sum_{kl} C_{kl}^{\tau} a_k^{+} a_l^{+} \tag{9.107}$$

*This contrasts with our convention of using capital letters for bosons; S_{+} (9.104) and D_{μ}^{+} (9.105) are collective fermion pairs. On the other side, the operators s^{+} and d^{+} in Eq. (9.119) are pure bosons.

and apply one of the methods discussed in the foregoing sections to construct a boson representation. Since the particle number is conserved, we now have only terms that conserve the boson number. In the Dyson method, for instance, we get a finite Hamiltonian of fourth order [see Eq. (9.71)]:

$$H_B = \sum_\tau \Omega_\tau B_\tau^+ B_\tau + \frac{1}{4} \sum_{\tau_1\tau_2\tau_3\tau_4} W_{\tau_1\tau_2\tau_3\tau_4} B_{\tau_1}^+ B_{\tau_2}^+ B_{\tau_3} B_{\tau_4}. \tag{9.108}$$

The coefficients C_{kl}^τ are now the *pp*-TDA amplitudes, and the quantities Ω_τ are given by the corresponding energies. In the other methods we end up with similar Hamiltonians (perhaps infinite and Hermitian but *boson number conserving*).

The disadvantage of this method is that it can be applied microscopically only for very few bosons, for instance, in the vicinity of a magic nucleus with a really inert core. For nuclei far away from magic shells we need a large number of *pp*-bosons whose corresponding C_{kl}^τ coefficients are determined on the basis of a completely inert core. For a large number of active particles this need not be a very good basis, however, as we can easily imagine in considering the extreme case of a ^{16}O core for the description of nuclei in the ^{208}Pb region. Thus, for nuclei far from being magic, the *pp*-boson description might not be valid if the amplitudes are restricted to the inert core basis. We may, however, try to consider the amplitudes C_{kl}^τ as adjustable parameters to be determined, for example, by a variational calculation.

On the other hand, configurations consisting of a large number of bosons can only be treated by group theoretical methods. So far, one has investigated only phenomenological models. One of this models is the *interacting boson model* ($SU(6)$-model) of Arima and Iachello [AI 75a and b]. It includes two types of bosons, an s-boson corresponding to a Cooper pair of Eq. (9.104), and five d-bosons corresponding to the five quadrupole pairs in Eq. (9.105). The Hermitian Hamiltonian has only boson number conserving terms up to fourth order [see Eq. (9.119)]:

$$s^+s, \quad d^+d, \quad d^+d^+dd, \quad s^+s^+ss, \quad d^+d^+ss, \quad d^+d^+ds \quad + \text{h.c.}$$

Neglecting the fourth-order terms which describe the interaction between the bosons, we have an unperturbed ground state as a product of $A/2$ s-bosons and excited states, where some of the s-bosons are replaced by a d-bosons. Originally, the model was introduced on purely phenomenological grounds. For one j-shell with a pairing-plus-quadrupole interaction (4.141), however, it can be motivated from a microscopic point of view by mapping *pp*-fermion pairs onto a suitable boson representation [AOI 77]. We shall discuss a few applications of this model in the next section.

Another example for *pp*-bosons is the *Schwinger representation* [BM 78]. Working in a single-particle basis, where a_m^+ and a_i^+ create particles above and below the fermi surface, we can introduce two types of bosons: B-bosons B_{mi}^+, which correspond to the creation of a *pp*-pair $a_m^+ a_i^+$ (with one particle above and one below the fermi surface); and A-bosons A_{ij}^+, which correspond to the creation of *pp*-pair $a_i^+ a_j^+$ (with both particles below the fermi surface). Both types of bosons are ideal bosons with the commutation relations

$$[B_{mi}, B_{m'i'}^+] = \delta_{mm'}\delta_{ii'}, \qquad [A_{ij}, A_{i'j'}^+] = \delta_{ii'}\delta_{jj'} - \delta_{ij'}\delta_{ji'}. \tag{9.109}$$

For the description of even systems we need only the density operators $a_k^+ a_l$. They

are mapped in the following way.

$$\begin{aligned} a_{m'}^{+}a_m &\to (a_{m'}^{+}a_m)_B = \sum_i B_{m'i}^{+}B_{mi}, \\ a_{i'}^{+}a_i &\to (a_{i'}^{+}a_i)_B = \sum_j A_{i'j}^{+}A_{ij}, \\ a_m^{+}a_i &\to (a_m^{+}a_i)_B = \sum_j B_{mj}^{+}A_{ij}, \\ a_i^{+}a_m &\to (a_i^{+}a_m)_B = (a_m^{+}a_i)_B^{+}. \end{aligned} \tag{9.110}$$

This mapping conserves the corresponding commutation relations for the density operators $a_k^{+}a_l$, Hermiticity, and furthermore ends up with finite expressions in the boson space. Starting from the boson vacuum $|-)$,

$$A_{ij}|-) = B_{mj}|-) = 0, \tag{9.111}$$

which corresponds to the bare vacuum in the fermion space we can construct the HF ground state $|\Phi_0)$ in the boson space by the conditions

$$(a_{m'}^{+}a_m)_B|\Phi_0) = 0, \qquad [\delta_{ii'} - (a_{i'}^{+}a_i)_B]|\Phi_0) = 0. \tag{9.112}$$

They are satisfied by

$$|\Phi_0) = \frac{1}{\sqrt{N!}} \det A^{+}|-) = \frac{1}{\sqrt{N!}} \begin{vmatrix} A_{11}^{+} \dots A_{N1}^{+} \\ \vdots \quad \vdots \\ A_{1N}^{+} \dots A_{NN} \end{vmatrix} |-). \tag{9.113}$$

Excited *ph*-states are constructed by successive applications of the operators $(a_m^{+}a_i)_B$. From Eq. (9.110), we see that the effect of such an operation is to replace the ith column of the operators A_{ij}^{+} by a column of B_{mj}^{+} in the determinant (9.113).

This boson representation is a straightforward generalization of the *Schwinger representation* (9.11) in the angular momentum case. A problem connected with this type of representations is the fact that we have introduced too many boson operators.

The state $|\Phi_0)$ and the many-particle–many-hole states built from it form a basis in the physical subspace of the bosons. We can show that the operator

$$L_{ii'} = \sum_j A_{ji}^{+}A_{ji'} + \sum_m B_{mi}^{+}B_{mi'} = (A^{+}A + B^{+}B)_{ii'} \tag{9.114}$$

annihilates all these states for $i \neq i'$ and leaves them unchanged for $i = i'$; that is, we have the constraint

$$L_{ii'} = (A^{+}A + B^{+}B)_{ii'} = \delta_{ii'} \tag{9.115}$$

in the physical subspace. It corresponds to the constraint (9.14) in the angular momentum case. Introducing the boson operators A and B enlarges the boson space drastically. As a result, we get a new class of additional spurious solutions violating the condition (9.115).

In principle, we can use the constraint (9.115) to eliminate the operators A_{ij}, A_{ij}^{+}. This is done formally by writing

$$A \sim \sqrt{1 - B^{+}B}\,. \tag{9.116}$$

Inserting the expression into Eq. (9.110), we again end up with the Holstein–Primakoff expression (9.4) in the physical subspace. A detailed analysis shows that the elimination (9.116) corresponds to a canonical transformation within the boson space [BM 78].

In a similar way we can eliminate the s-boson in the $SU(6)$ model of Arima and Iachello. We then end up with a Holstein–Primakoff representation of this model containing only d-bosons with square root factors of the type (9.116). In this form the $SU(6)$ model has been first investigated by Janssen et al. [JJD 74].

So far we have used only the particle representation to discuss the Schwinger bosons. There have been also attempts to introduce a general Schwinger representation based on quasi-particles [YN 76, MNY 77, FYN 77, Fu 77].

9.2.8 Applications

Several groups have applied the method of boson expansions to the description of transitional nuclei, starting with weakly anharmonic vibrators through strong anharmonicities, to the rotational limit in well deformed nuclei. Basically, we have to distinguish between two types of work, namely *phenomenological approaches*, which parametrize boson Hamiltonians of a simple structure and *microscopic approaches*, where the parameters in the boson Hamiltonian are derived from an underlying fermion many-body Hamiltonian.

9.2.8.1. Phenomenological and Interacting Boson Models. We can express the Hamiltonian by phenomenological boson operators whose physical structure need not be specified in detail. According to the number of different bosons and the order to which we go, it contains a number of parameters h_{lm}. An example is provided by the quadrupole bosons d_μ^+ up to fourth order. (The word quadrupole in this context only means that these operators are spherical tensors of rank 2. It does not specify the radial dependence of a corresponding fermion operator.)

$$\begin{aligned} H = {} & h_{00} + h_{11}[d^+ d]_0 + h_{20}[d^+ d^+]_0 + h_{30}[[d^+ d^+]_2 d^+]_0 + h_{21}[[d^+ d^+]_2 d]_0 \\ & + h_{40}[d^+ d^+]_0 [d^+ d^+]_0 + h_{31}[d^+ d^+]_0 [d^+ d]_0 \\ & + \sum_{L=0,2,4} h_{22}^L [[d^+ d^+]_L [dd]_L]_0 . \end{aligned} \tag{9.117}$$

The brackets indicate the coupling of the angular momenta. The diagonalization of such an operator is a nontrivial problem. One usually calculates its matrix elements between multi-boson states and diagonalizes the corresponding matrix. However, the size of this matrix can only be kept within reasonable limits by using the group theoretical properties of the underlying Hamiltonian. For instance, the quadrupole bosons obey the $SU(5)$ symmetry, which has been studied in great detail in the literature [WHG 66, KPW 68, WP 68, KT 71].

The diagonalization of the phenomenological Hamiltonian (9.117) gives a spectrum that depends on the constants h_{lm}. In the calculation of *transition probabilities*, we need a boson representation of the transition operator containing additional constants. The fitting of all these parameters to experimental data allows a test, to see if the observed physical phenomena can be described by the corresponding bosons. If they can, we

also get information about the Hamiltonian, such as statements concerning the parameters. On the other hand, as we have no microscopic derivation, we have to be very careful, because if we use enough parameters sometimes quite different physical phenomena may be fitted.

The simplest version of such phenomenological approaches uses only the five quadrupole bosons and a fourth-order coupling term diagonal in the boson number. In such cases, we can fit the highest angular momentum members of the different quadrupole multiplets reasonably well [BTK 65, DDK 70, HR 74, IA 74]. To describe the other members we also need a more general ansatz, like (9.117) [KS 62, GG 71, HL 72a].

Starting from the boson operators B, B^+, we can define corresponding coordinates Q and momenta P [see, for instance, Eq. (8.95)] and formally divide the Hamiltonian into a potential and kinetic energy part

$$H(P,Q)=\frac{1}{2M}P^2+V(Q). \tag{9.118}$$

The requirement that the "mass parameter" depends on neither P nor Q leads to simple relations between the parameters of the boson Hamiltonian [Ho 71a]. In the quadrupole model, the potentials in the internal frame depend on the deformation parameters (see Eq. (1.13)). Figure 9.4 shows such a potential curve as a function of β and the corresponding spectrum in the nucleus ^{76}Se. The constants of the boson Hamiltonian were obtained using a fit of four phenomenological parameters.

One is also able to describe the transition from a spherical vibrator spectrum, through anharmonicities to spectra with a rotational band struc-

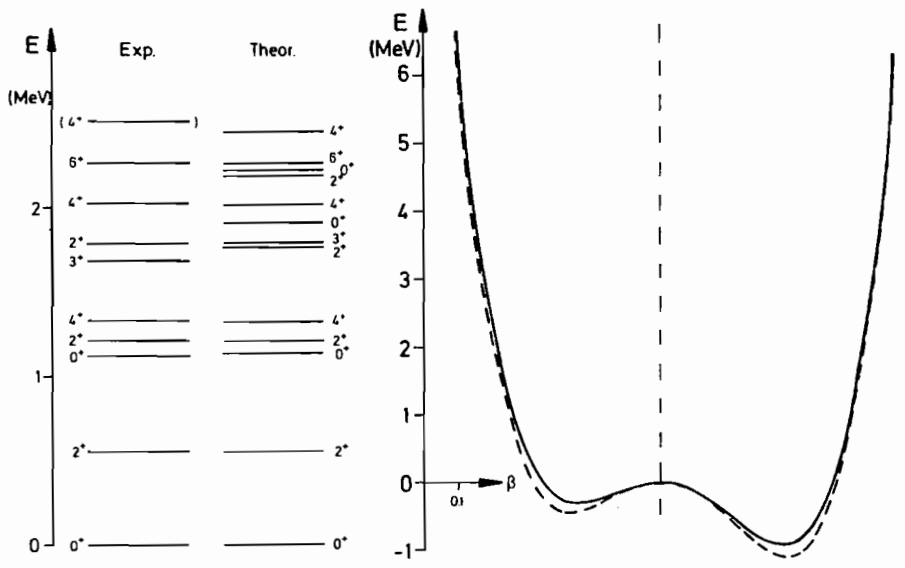


Figure 9.4. The experimental energy levels of ^{76}Se compared to the eigenvalues of a phenomenological boson Hamiltonian. The full curve shows the corresponding potential energy in the intrinsic $\beta-\gamma$ frame at $\gamma=0^0$. The dashed curve is the potential arising from a microscopic calculation. (From [LH 75].)

ture using an appropriate choice of the parameters in the Hamiltonian [Ho 70, GMG 69, 70, DB 70, GG 71, Ho 71a]. A rotational band structure develops if the potential $V(Q)$ has a well-pronounced deformed minimum (see Fig. 9.5). However, for such cases a large number of bosons (up to 30) is needed. Besides the tremendous numerical effort brought about by the components with high boson numbers, also serious doubts from a physical point of view may exist, since they strongly violate the Pauli principle and we do not know if the phenomenological Hamiltonian really takes care of this fact.

Over the years, a large number of phenomenologically based models have been developed whose diagonalizations are considerably simplified by the use of group theoretical techniques (see, for instance, [HNL 70, DDK 70, VNR 75, RD 76, RGB 77]). Of these, probably the best known is the *interacting boson model* ($SU(6)$-model), which was first proposed by Janssen et al. [JJD 74] and later on much investigated by Arima and Jachello [AI 75a, b]. Its six basic blocks are five quadrupole bosons d_μ^+ ($\mu=-2,\ldots,+2$) and one pairing boson s^+. The boson number is conserved, and it is a model having *pp*-bosons (a boson representation of the Schwinger type) as discussed in Section 9.2.7. The Hamiltonian has the form

$$H=\epsilon_s s^+ s+\epsilon_d[d^+d]_0+\sum_{L=0,2,4} c_L[[d^+d^+]_L[dd]_L]_0+u_0 s^+ s^+ ss$$
$$+\{v_2[[d^+d^+]_2[ds]_2]_0+v_0[d^+d^+]_0 ss+u_2[[d^+s^+]_2[ds]_2]_0+\text{h.c.}\}. \tag{9.119}$$

It contains eight parameters $\epsilon=\epsilon_d-\epsilon_s, c_0, c_2, c_4, u_0, u_2, v_0, v_2$ which are fitted to experimental data. In addition, we need two further constants to define the $E2$-transition operator

$$T_{2\mu}=q_0([d^+s]_{2\mu}+\text{h.c.})+q_1[d^+d]_{2\mu}. \tag{9.120}$$

In the case of a vanishing interaction ($c_L=v_L=u_L=0$), the ground state consists only of s-bosons:

$$|0)=\frac{1}{\sqrt{n!}}(s^+)^n|-),$$

in which $|-)$ is the bare vacuum and $2n$ the number of nucleons. Successively replacing one or several s-bosons by d-bosons, we obtain quadrupole excitations in the same nucleus with the excitation energies $\epsilon=\epsilon_d-\epsilon_s$,

$$|\mu)=\frac{1}{\sqrt{(n-1)!}}d_\mu^+(s^+)^{n-1}|-).$$

The unperturbed spectrum is therefore a purely harmonic quadrupole spectrum, as shown in Fig. 1.5.

Adding further s-bosons, we obtain states in the neighboring even nuclei with particle numbers $2n+2, 2n+4,\ldots$, etc., as was discussed in the seniority model (see Sec. 8.2).

The boson–boson interaction induces correlations. The Hamiltonian (9.119) then has to be diagonalized in a multi-boson basis containing states with n_s s-bosons and $n_d = n - n_s$ d-bosons $|n_s, n_d, \alpha)$. (α contains all remaining quantum numbers needed for the unique specification of the states.) The evaluation of the corresponding matrix elements is greatly simplified by group theoretical methods.

The relation

$$n = \hat{n}_s + \hat{n}_d = s^+ s + \sum_\mu d_\mu^+ d_\mu$$

can be used to *eliminate the operators s and* s^+. For a fixed value of n the basis states are completely characterized by the quantum numbers of the d-bosons, and we obtain the proper matrix elements of the Hamiltonian if we replace the operators s by

$$s \to \sqrt{n - \sum_\mu d_\mu^+ d_\mu} \ . \tag{9.121}$$

For example, we have

$$s|n_s = n - n_d, n_d, \alpha) = \sqrt{n_s}\ |n - n_d - 1, n_d, \alpha)$$

$$= \sqrt{n - \sum_\mu d_\mu^+ d_\mu}\ |n - n_d - 1, n_d, \alpha).$$

Applying these rules to the operator (9.119), we end up with a Hamiltonian containing only the quadrupole bosons d_μ^+. However, it no longer conserves the boson number, and by expanding the square roots in Eq. (9.121) we obtain an infinite series of terms $d^+ d \ldots$. This is the Holstein–Primakoff version of the $SU(6)$-model [JJD 74]. The introduction of the s-boson is therefore a very elegant tool for treating the square root factors introduced by the Pauli principle and ending up with a finite boson representation. We emphasize, however, that the s-boson does not introduce a new degree of freedom for a fixed particle number. It is only used to construct the unperturbed ground state. The model contains no 0^+ pairing vibrations of the $2p-2h$ type as discussed in Section 8.3.5. On the other hand, the introduction of the s-boson allows us to treat states in the adjacent $A+2, A+4, \ldots$ nuclei.

Depending on the actual values of the parameters, the model has several *limits*, which correspond to various subgroups of $SU(6)$ [CCF 79] and are sometimes analytically soluble. They also represent very important physical mechanisms in actual nuclei:

(i) For $c_L = v_L = 0$, $u_2 = 0$ (i.e., without coupling to d-bosons), we obtain the *seniority scheme* in a single j-shell ($SU(2)$) as discussed in Section 6.2:

$$H = \epsilon_s s^+ s + u_0 s^+ s^+ ss = -GS_+ S_- ,$$

with the quasi-spin operators

$$S_+ = \sqrt{\Omega} \cdot s^+ \sqrt{1 - \frac{s^+ s}{\Omega}}$$

and

$$u_0 = -G, \qquad \epsilon_s = G \cdot \Omega.$$

(ii) For weak interaction $u_L = 0$; $v_L, c_L \ll \epsilon_d - \epsilon_s$ we have the *vibrational limit* [AI 76] with the symmetry $SU(5)$. The s-bosons play no role in this limit, and in the extreme case we have only interacting quadrupole bosons (see also [BTK 65, DDK 70, IA 74]) with the Hamiltonian

$$H = \sum_{\mu} \epsilon_d d_\mu^+ d_\mu + \sum_{L=0,2,4} c_L [[d^+ d^+]_L [dd]_L]_0. \qquad (9.122)$$

Its eigenvalues and transition rates can be given in analytic form.

(iii) If certain relations between the parameters of the model are fullfilled, we can rewrite the Hamiltonian (9.119) in the form

$$H = \epsilon \sum_{\mu} d_\mu^+ d_\mu - \kappa' [L \cdot L]_0 - \kappa [Q \cdot Q]_0, \qquad (9.123)$$

with the parameters $\epsilon, \kappa, \kappa'$, the three angular momenta

$$L_\mu = \sqrt{10}\, [d^+ d]_{1\mu},$$

and the five quadrupole operators

$$Q_\mu = d_\mu^+ s + s^+ d_\mu - \frac{\sqrt{7}}{2} [d^+ d]_{2\mu}.$$

For $\epsilon = \kappa' = 0$, we obtain the *rotational limit* [AI 78a] based on the group $SU(3)$. It is an extension of the Elliot model [El 58], which was widely used for fermions and whose group theoretical properties are discussed in the literature [Ha 68, Ve 68, JDJ 75]. It has the spectrum of an axially symmetric rigid rotor (with degenerate 2_β^+ and 2_γ^+ levels)

$$E(\lambda, \mu, K, L, M) = \tfrac{3}{4}(\kappa - \kappa') L \cdot (L+1) - \kappa\big(\lambda^2 + \mu^2 + \lambda\mu + 3(\lambda + \mu)\big),$$

where (λ, μ, K, L, M) are the quantum numbers of the group $SU(3)$.

(iv) At the end of each major shell, another subgroup of the $SU(6)$, the $O(6)$ seem to play an important role for the classification of the spectra [AI 78b, CC 78, CBP 78]. Again, the eigenvalues can be given in analytic form. In a certain limit they are identical to those of a deformed, completely γ-soft oscillator, the Wilets–Jean model [WJ 56, Me 78].

Since the $SU(6)$ model incorporates so many important physical effects and also has a rather large number of free parameters (up to eleven), we can get excellent agreement with many experimental spectra in spherical, and weakly and strongly deformed nuclei (see also [Ia 77, AOI 78, Ia 78]). We can also give a microscopic justification for the case of a pairing-plus-quadrupole interaction within a single j-shell [OAI 78]. A full microscopic derivation, which also includes *single-particle degrees* of freedom is, however, still missing.

9.2.8.2. Microscopic Calculations. Although the phenomenological boson models have had great success in fitting experimental spectra and transition probabilities, we can never be sure that they do not simulate quite different physical mechanisms having nothing to do with bosons, by a suitable choice of parameters. It is therefore crucial in this kind of theory to demonstrate that we can, in principle, derive the parameters entering the Hamiltonian from a microscopic point of view.

Lee and Holzwarth [LH 75] have used the Marumori method (Sec. 9.2.4) with collective bosons for the description of anharmonic nuclei with a rather pure quadrupole structure. They expanded the Hamiltonian of a surface delta interaction [see Eq. (4.117)] up to fourth order in TDA bosons with angular momentum two and found that the resulting expansion coefficients h_{lm} in Eq. (9.117) are generally in good agreement with the parameters determined by fitting the experimental levels [HL 72a]. Figure (9.4) shows the corresponding microscopically determined boson potential $V(Q)$ (dashed line).

B. Sørenson used the Belyaev–Zelevinskii method described in Section 9.2.3.2. In several papers [Sø 67, 68a+b, 69, 70] he investigated pairing modes near closed shells and quadrupole modes in transitional nuclei. He used a slightly modified pairing-plus-quadrupole interaction [see Eq. (4.141)] and expanded the Hamiltonian to fourth order. He successfully reproduced, at least semiquantitatively, the main features of such nuclei. However, he had difficulties in cases where the coupling to non-collective two-quasi-particle states are important.

More complete investigations based on the same method have been

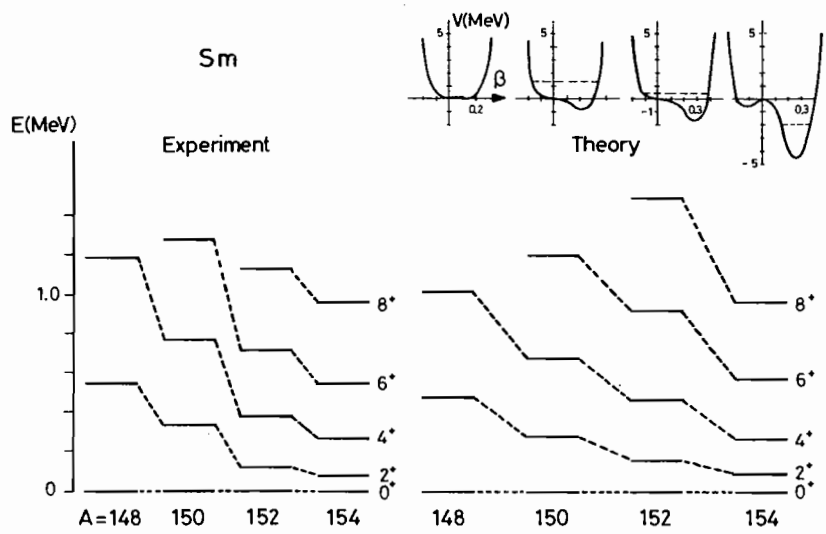


Figure 9.5. The phase transition spherical—deformed in the Samarium nuclei. The structure of the spectrum changes from a vibrator to a rotator. Also shown are corresponding potentials of the boson Hamiltonian as a function of the deformation parameter. (From [KT 76].)

carried out by Kishimoto and Tamura [KT 72, 76]. In addition to the pairing-plus-quadrupole interaction they used a quadrupole pairing force [see Eq. (4.134)] and expanded the Hamiltonian to sixth order in the collective branch of ($I=2$)-bosons. The coupling to the non-collective branches turned out to be very important in many nuclei. It was taken into account by a projector formalism in the sense of Feshbach [Fe 58, 62], which ends up with an effective Hamiltonian in the collective subspace. Thus the parameters h_{lm} had to be replaced by energy-dependent renormalized parameters $\tilde{h}_{lm}$. The results of these complicated calculations show extremely good agreement with experimental data not only for vibrational, but also for transitional and even purely rotational nuclei. Figure 9.5 gives an example for the transition from spherical to deformed shapes in the Sm region. Table 9.1 gives the corresponding electric properties.

Table 9.1. Spectroscopic quadrupole moments and $BE2$ values of the first excited 2^+ state calculated with the boson expansion method and compared with experimental data (from [TK 73])

	$Q\,[e\cdot b]$		$BE2(2^+\to 0^+)\,[e^2\cdot b^2]$	
	Experiment	Theory	Experiment	Theory
^{148}Sm	-0.97 ± 0.27	-0.91	0.141 ± 0.005	0.152
^{150}Sm	-1.31 ± 0.19	-1.17	0.274 ± 0.006	0.270
^{152}Sm	-1.73 ± 0.20	-1.72	0.666 ± 0.014	0.697
^{154}Sm	——	-1.91	0.922 ± 0.040	0.884

9.3 Odd Mass Nuclei and Particle Vibration Coupling

Until now we have considered only systems with an even number of particles. Systems with odd particle number also, of course, have collective states. Historically, one of the most famous examples is the septuplet of ^{209}Bi, where the $1h_{9/2}$ proton of ^{209}Bi couples to the 3^- vibration at 2.61 MeV of the ^{208}Pb core. The proton perturbes this vibration only very slightly, giving rise to a very small splitting of the seven different spin states (see Fig. 9.6). These states are very often treated in the vibration-

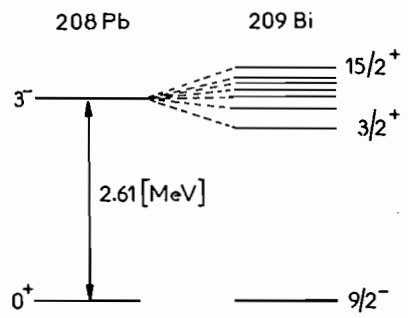


Figure 9.6. Schematic comparison of the experimental spectra of ^{208}Pb and ^{209}Bi.

particle coupling (VPC) model [BM 75]. It turns out that this is closely connected to the boson expansion in odd systems.

9.3.1 Boson Expansion for Odd Mass Systems

All different kinds of boson theories considered in the last section can be generalized to odd mass systems [Ya 65, DJ 73, Ma 73, Ma 74, Pr 74, RS 77a, BM 78]. Since the principle is not very different from the even case, we only want to sketch briefly the Belyaev–Zelevinskii and Dyson methods.

In the Belyaev–Zelevinskii method we now consider, in addition to the fermion pair operators $b_\mu^+, a_{\mu\nu}$ [Eqs. (9.24) and (9.25)], the fermion operators α_k^+, and map these operators into an enlarged space, namely the direct product of the usual boson space of the B_μ^+ and a space of *one* ideal fermion β_k^+ which commutes with the bosons.

$$\left[B_\mu, \beta_k^+\right] = \left[B_\mu^+, \beta_k^+\right] = 0. \tag{9.124}$$

Since there is only one ideal fermion at a time, the operators β_k obey the relations [Ok 74]

$$\beta_k \beta_l = \beta_k^+ \beta_l^+ = 0, \qquad \beta_k \beta_l^+ = \delta_{kl}. \tag{9.125}$$

The vacuum projector in this space is given by

$$|0)(0| =: \exp\left(-\sum_\mu B_\mu^+ B_\mu\right) : \left(1 - \sum_k \beta_k^+ \beta_k\right), \tag{9.126}$$

which again means

$$B_\mu |0) = \beta_k |0) = 0. \tag{9.127}$$

The basis states in the product space are therefore given by

$$|k, N) = \mathcal{N}_B \cdot \beta_k^+ B_{\mu_1}^+ \ldots B_{\mu_N}^+ |0) \qquad N = 0, 1, \ldots . \tag{9.128}$$

In addition to the algebra (9.27) of the operators b_μ^+ and $a_{\mu\nu}$, we now also have to consider commutation relations between single-fermion and fermion pair operators:

$$\begin{aligned} &\{\alpha_k, \alpha_l\} = \{\alpha_k^+, \alpha_l^+\} = 0, \qquad \{\alpha_k, \alpha_l^+\} = \delta_{kl}, \\ &\left[\alpha_k, b_\mu\right] = 0, \qquad \left[\alpha_k, b_\mu^+\right] = \sum_l C_{kl}^\mu \alpha_l^+, \\ &\left[\alpha_k, a_{\mu\nu}\right] = \sum_l \Gamma_{\nu l}^{\mu k} \alpha_l, \end{aligned} \tag{9.129}$$

with $\Gamma_{\nu l}^{\mu k}$ defined by Eq. (9.29).

We now proceed as in Section 9.2.3.1 and make the following ansatz for the boson image of the different operators (arguments for the specific choice of the ansatz are given in [Ma 74]); here we want to content ourselves with the fact that the commutation rules are satisfied order by

order*:

$$(a_{\mu\nu})_B = \sum_{\rho\sigma} \Gamma^{\mu\rho}_{\nu\sigma} B_\rho^+ B_\sigma + \sum_{kl} \Gamma^{\mu k}_{\nu l} \beta_k^+ \beta_l,$$

$$(b_\mu^+)_B = B_\mu^+ - \frac{1}{4} \sum_{\nu\rho\sigma} \Gamma^{\nu\rho}_{\mu\sigma} B_\nu^+ B_\rho^+ B_\sigma + \cdots - \frac{1}{2} \sum_{\nu kl} \Gamma^{\nu k}_{\mu l} B_\nu^+ \beta_k^+ \beta_l + \cdots, \tag{9.130}$$

$$(\alpha_k^+)_B = \beta_k^+ - \frac{1}{2} \sum_{\rho\sigma l} \Gamma^{\rho l}_{\sigma k} B_\rho^+ B_\sigma \beta_l^+ + \cdots + \sum_{\rho l} C^{\rho *}_{kl} B_\rho^+ \beta_l.$$

With these formulas we can construct the boson image of the Hamiltonian operator (9.32).

9.3.2 Derivation of the Particle Vibration Coupling (Bohr) Hamiltonian

Insertion of (9.130) into the Hamiltonian (9.32), (9.33) yields to first order in the Γ's if we neglect for the moment ground state correlations, that is, H^{40}†:

$$H = \sum_\mu \Omega_\mu B_\mu^+ B_\mu + \sum_k E_k \beta_k^+ \beta_k + \cdots + H_{\text{coupl}},$$

$$H_{\text{coupl}} = \sum_{\mu,k,l} \frac{1}{2} (\gamma^\mu_{kl} B_\mu^+ \beta_k^+ \beta_l + \text{h.c.}) - \frac{1}{2} \sum_{\substack{\mu\nu\lambda \\ kl}} V^{22}_{\mu\nu} \cdot (\Gamma^{\lambda l *}_{\nu k} B_\mu^+ B_\lambda + \Gamma^{\lambda k}_{\mu l} B_\lambda^+ B_\nu) \beta_k^+ \beta_l, \tag{9.131}$$

with

$$\gamma^\mu_{kl} = 2 \sum_{rs} H^{31}_{rskl} C^{\mu *}_{rs}, \tag{9.132}$$

where C^μ_{kl} are the TDA amplitudes and Ω_μ the corresponding energies. In Eq. (9.131) we neglected B^+B terms containing two Γs, since they are usually smaller than those with only one Γ. The first term of the particle vibration coupling H_{coupl} is known as the Bohr coupling Hamiltonian [BM

* In the non-collective case we can, in analogy to Eq. (9.103), sum up the infinite series for $(b^+_{kl})_B$ and $(\alpha_l^+)_B$:

$$(a_{kl})_B = \sum_r B_{lr}^+ B_{kr} + \beta_l^+ \beta_k,$$

$$(b^+_{kl})_B = (B^+ \sqrt{1 - B^+ B})_{kl} + \sum_{pq} \left[(\sqrt{1 - B^+ B})_{pq}, B^+_{kl} \right] \beta_p^+ \beta_q,$$

$$(\alpha_k^+)_B = \sum_l (\sqrt{1 - B^+ B})_{lk} \beta_l^+ + B^+_{kl} \beta_l.$$

† Since we neglect H^{40}, terms like B^+B^+ and BB are missing. We could take them into account by transforming to RPA bosons with (see Appendix E.5)

$$B_\mu = \sum_{\substack{mi \\ \rho}} C^{\mu *}_{mi} (X^\rho_{mi} Q_\rho + Y^{\rho *}_{mi} Q_\rho^+).$$

75], since it has also been derived within the liquid drop model (see Sec. 9.3.4). The second term in H_{coupl} comes from a systematic and microscopic treatment of the boson expansion and corrects for the Pauli principle, as we shall see. The matrix element H^{31} can be calculated in the canonical basis (see Chap. 7) from Eq. (E.25) for systems with time reversal invariance:

$$H^{31}_{rskl} = \frac{1}{3}\cdot\frac{1}{4}\{-\bar{v}_{rk\bar{s}l}(\eta^+_{rs}\xi^+_{kl}+\eta^-_{rs}\xi^-_{kl}) + \bar{v}_{sk\bar{r}l}(\eta^+_{rs}\xi^+_{kl}-\eta^-_{rs}\xi^-_{kl}) + \bar{v}_{rs\bar{k}l}(\xi^+_{rs}\eta^+_{kl}-\xi^-_{rs}\eta^-_{kl})\}, \tag{9.133}$$

where the occupation factors ξ, η are defined in Eq. (8.202). In the following we investigate, in more detail, cases without pairing correlations. The quasi-particle operators β_k^+ then have to be replaced by particle operators (see Chap. 2) c_m^+ or hole operators c_i (we use the indices m, n for particles and i, j for holes). Instead of quasi-particle energies we find ϵ_m or $-\epsilon_i$ (where the energy scale $\epsilon_F = 0$ is used). We obtain two types of vertices, because in our quasi-particle formalism we can represent a *ph*-vibration (indices $\nu, \nu', \ldots$) or a *pp* (*hh*) vibration (indices $\tau, \tau', \ldots$). Neglecting pairing correlations, therefore, if we consider a particle coupled to a $1p-1h$ state (note that the original C^{μ}_{rs} are antisymmetric) we obtain for the first term of H_{coupl}:

$$H_{\text{coupl}} = \frac{1}{3}\sum_{\nu}\sum_{mn}\sum_{m'i'} C^{\nu *}_{m'i'}\bar{v}_{m'mi'n}B_\nu^+ c_m^+ c_n + \text{h.c.} - \frac{1}{3}\frac{1}{2}\sum_{\tau}\sum_{jn}\sum_{m'n'} C^{\tau *}_{m'n'}\bar{v}_{m'n'jn}B_\tau^+ c_j c_n + \text{h.c.} \tag{9.134}$$

The origin of the factor $\frac{1}{3}$ comes from the fact that our fermion–boson basis is three times as large as our original basis, having included two possibilities of coupling a particle to a *ph*-vibration and one possibility of coupling a hole to a *pp* vibration (see also below). Using our graphical rules (Chap. 8 and Appendix F), we can represent the two vertices γ^{ν}_{mn} and γ^{τ}_{jn} as shown in Fig. 9.7. In order to facilitate the discussion somewhat we want to disregard the possibility of *pp* vibrations for the moment, that is, the second term of (9.134) will drop out. Of course, we should rederive our boson expansion from the beginning under this assumption; we do not want to do this here, but rather state that we find that the factor $\frac{1}{3}$ in Eq.



Figure 9.7. Vertices for the coupling of a particle to a particle plus *ph* vibration and the coupling of a particle to a hole plus *pp* vibration.

(9.134) is changed to $\frac{1}{2}$, that is, *

$$H_{\text{coupl}} = \frac{1}{2} \sum_{\nu} (\gamma^{\nu}_{mn} B_{\nu}^{+} c_m^{+} c_n + \text{h.c.}),$$
$$\gamma^{\nu}_{kl} = \sum_{mi} C^{\nu *}_{m'i'} \bar{v}_{m'ki'l}. \tag{9.135}$$

In order to treat collective vibrations of odd nuclei, we must formally diagonalize the Hamiltonian (9.131) in the multi-phonon–one-particle basis (9.128). Of course, in practice again, often only one phonon (for instance, the quadrupole or octupole phonon) will be taken into account. The more anharmonic the underlying vibrator is, the more multi-phonon components we need for the diagonalization of the Hamiltonian (9.131). This procedure has been pursued in a somewhat more phenomenological version by Alaga and collaborators (see, eg., [Al 67, APS 75] and references therein). As we have seen in the case of even systems, the boson description works quite well even for rotational nuclei. The Hamiltonian (9.131) is therefore, in principle, capable of describing all the effects occurring if an odd particle is coupled to a rotor as, for instance, rotational alignment or the pattern of triaxial shapes. Few attempts in this direction have been made [AP 76, PVD 77]. Applications of this Hamiltonian [(9.131), (9.134)] have, however, been made mainly for rather harmonic vibrators, to which we shall now turn.

9.3.3 Particle Vibration Coupling (Perturbation Theory)

As we said previously, the Hamiltonian (9.131) is suitable for treating collective vibrations in odd nuclei. In the simplest case we consider the neighboring even–even nucleus to be a harmonic vibrator, and the first collective state is then a vibration particle state. Furthermore it sometimes happens that the vibration particle coupling (VPC) is so weak that H_{coupl} can be treated in perturbation theory (weak coupling situation). If we want to calculate, for example, the energy shift of the vibration particle state due to the coupling terms, we obtain to second order:

$$\Delta E = \Delta E_1 + \Delta E_2 = (p\mu|H_{\text{coupl}}|p\mu) + \sum_{\substack{kN \\ \neq p\mu}} \frac{(p\mu|H_{\text{coupl}}|kN)(kN|H_{\text{coupl}}|p\mu)}{E_{p\mu} - E_{kN}}, \tag{9.136}$$

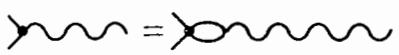
where $|p\mu) = \beta_p^{+} B_{\mu}^{+}|0)$ is the unperturbed configuration and the $|kN)$ are defined in Eq. (9.128). In first order only the second term of H_{coupl} contributes, and we obtain from (9.33), (9.26), and (9.131):

$$(p\mu|H_{\text{coupl}}|p\mu) = -\frac{1}{2} \sum_{\nu} \left(V^{22}_{\mu\nu} \Gamma^{\nu p *}_{\nu p} + \text{c.c.} \right)$$
$$= -\frac{1}{4} \sum_{klq} \left(H^{22}_{klpq} C^{\mu *}_{kl} C^{\mu}_{pq} + \text{c.c.} \right), \tag{9.137}$$

with H^{22} defined by Eq. (8.200 and 8.201).

*The interaction that we use here is formally the bare one, as in all boson expansion theories we have used so far, since we started from the original fermion Hamiltonian. It can be shown, however (see Appendix F), that the bare interaction in the vertex (9.135) can be replaced by an effective *ph* hole force, which is not necessarily antisymmetric.

As we already mentioned, there are several other boson expansion techniques that can also be applied to odd systems (but which will not be presented here). A somewhat different but related method is *nuclear field theory* (NFT) (see also below), which has been introduced by Bohr and Mottelson [Mo 67, BM 75] and worked out in detail by Broglia et al. [BBB 77 and references therein] and has been applied to odd systems. In order to facilitate the comparison of our present treatment with NFT we introduce an additional rule: Whenever the two indices of a TDA amplitude like C^{μ}_{pq} in Eq. (9.137) do not go to the same matrix element and are not summed over, we shall have to use the TDA equation (8.10)



in order to make this type of amplitudes change to ones in which both indices are on the same $\bar{v}$ and are summed over. Applying this rule to the first-order expression (9.137) and going over to the HF limit [see Eq. (8.202)], we obtain, if we restrict ourselves to *ph* vibrations only, with (9.135)

$$\Delta E_1 = -\sum_i \frac{|\gamma^{\mu}_{ip}|^2}{\Omega_{\mu} - \epsilon_p + \epsilon_i}. \tag{9.138}$$

In second order, only the first term of H_{coupl} contributes, and we obtain two terms coming from intermediate states $|k, N)$ containing zero ($N=0$) and two ($N=2$) bosons. Again in the HF limit, we have*

$$\Delta E_2 = \sum_n \frac{|\gamma^{\mu}_{pn}|^2}{\Omega_{\mu} + \epsilon_p - \epsilon_n} + \sum_n \frac{|\gamma^{\mu}_{np}|^2}{\epsilon_p - \epsilon_n - \Omega_{\mu}}. \tag{9.139}$$

According to our graphical rules (see Chap. 8 and Appendix F), the first and second term of ΔE_2 are represented by the graphs a and b, respectively, in Fig. 9.8 and ΔE_1 by graph c. The process shown in Fig. 9.8d clearly involves a matrix element of H^{40} (see Fig. 8.10), and we would have obtained it by taking into account H^{40} terms in Eq. (9.131) and transforming to RPA bosons, as previously mentioned. This quite lengthy procedure would have given a second first-order contribution corresponding to graph d. Neglecting the small RPA amplitudes, its analytic expression is given by

$$\sum_i \frac{|\gamma^{\mu}_{pi}|^2}{\Omega_{\mu} + \epsilon_p - \epsilon_i}. \tag{9.140}$$

The physical meaning of the different graphs is as follows.

*The strict application of perturbation theory (9.136) gives different weight factors for the various contributions than those we write down. [The first term in (9.139), for example, is multiplied by a factor $\frac{1}{4}$]. It is, however, quite a tricky point that in diagonalizing the second term of H_{coupl} in Eq. (9.131), that is, in summing its perturbation series to infinite order, additional terms like those given in (9.139) are generated (in addition to others, of course). Including them gives the factors (one) that we write down. Formally, then, this looks as if we had suppressed the factor $\frac{1}{2}$ in Eq. (9.135). It would go beyond the scope of this book to try to explain in detail how this comes about. This factor $\frac{1}{2}$ has, however, been a long-standing controversy in the literature [JDK 69, El 70] and we can make the following statements: Whenever a full diagonalization of the coupling term (9.131) is made, the factor $\frac{1}{2}$ must be kept; whenever we want to do only lowest-order perturbation theory this factor should be dropped. In any case, we should handle a perturbative treatment like, for instance, that of a collective boson Hamiltonian with great care, since the bosons, of course, contain an infinite number of interactions, making a consistent perturbative expansion in interactions and collective bosons quite a subtle question.

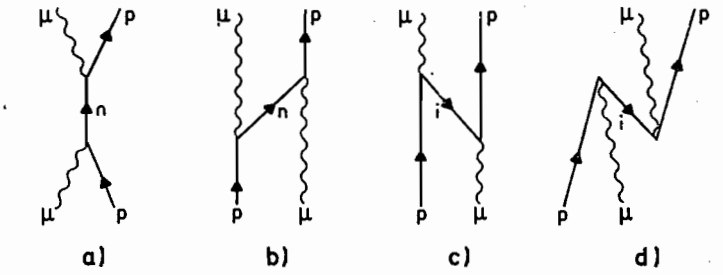


Figure 9.8. Four possible diagrams representing the second-order contribution to the particle—*ph* vibration coupling.

(a) The vibration and particle combine into a particle which afterwards disintegrates once again into a particle and a vibration.
(b) As the particle and vibration travel along, the particle excites a vibration and is itself scattered into another particle state which later on absorbs the incoming vibration.
(c) The incoming vibration disintegrates into a particle–hole pair, the hole of which later combines with the incoming particle, forming an outgoing vibration. Since this represents nothing but an exchange of the hole from one particle partner to another, this process takes into account the Pauli principle. We also see that this term comes from a higher boson term in H_{coupl}, where we know from the discussions for the even system, they represent corrections to the Pauli principle.
(d) This process is the same as the one in graph a, only the intermediate particle is changed to an intermediate hole. As we said, this process is due to correlations in the ground state.

The terms *a* and *b* have been taken into account by Kisslingler and Sorenson [KS 63], for example, but that the coupling to holes may also be quite important was pointed out for the first time by Hafele [Ha 67]. The relative importance of all these graphs for the particle 3^- phonon coupling in ^{209}Bi has been discussed by Mottelson [Mo 67].

9.3.4 The Nature of the Particle Vibration Coupling Vertex

Before we go on, let us briefly discuss in some detail the vertex γ^{μ}_{kl} (9.135), which is the main ingredient in all four processes shown in Fig. 9.8. For this purpose we want to write the vertex in the r-representation viz:

$$\gamma^{\mu}_{kl} = \int d^3r_1\, d^3r_2\, \varphi_k(\mathbf{r}_1)\gamma^{\mu}(\mathbf{r}_1, \mathbf{r}_2)\varphi_l^*(\mathbf{r}_2)$$
$$\gamma^{\mu}(\mathbf{r}_2, \mathbf{r}_4) = \sum_{mi} \int d^3r_1\, d^3r_3\, \bar{v}(\mathbf{r}_1\mathbf{r}_2\mathbf{r}_3\mathbf{r}_4)\varphi_m^*(\mathbf{r}_1)C^{\mu *}_{mi}\varphi_i(\mathbf{r}_3). \tag{9.141}$$

For the purpose of this qualitative discussion we approximate the force for the direct term by a pure Wigner δ-interaction (see Chap. 4), neglect the exchange, and obtain

$$\gamma^{\mu}(\mathbf{r}_1, \mathbf{r}_2) = \delta(\mathbf{r}_1 - \mathbf{r}_2)\gamma^{\mu}(\mathbf{r}_1), \qquad \gamma^{\mu}(\mathbf{r}) \propto \delta\rho^{\mu}(\mathbf{r}), \tag{9.142}$$

where $\delta\rho^{\mu}(\mathbf{r})$ is the transition density* discussed in Section 8.3.1. From the sum rule (see Sec. 8.7.5) and semiclassical [see Tassie model, Eq. (8.175)] considerations, we find the following form for the transition density expressed in terms of the spherical density ρ_0:

$$\delta\rho^{\lambda} \propto \lambda r^{\lambda-1} Y_{\lambda 0} \cdot \frac{d\rho_0}{dr}, \qquad \lambda \neq 0. \tag{9.143}$$

We see, therefore, that the particle vibration coupling takes place predominantly in the surface region of the nucleus. It is interesting to study the very close relationship to the liquid drop model (see [BM 75] and Chap. 1): As usual, in this model the equipotential surfaces of the nuclear potential $V(\mathbf{r})$, for a fixed value of $\mathbf{r}$, are supposed to be given for small deformations by Eq. (1.7) [Ha 74a]:

$$r_{\Omega} = r \cdot \left(1 + \sum_{\lambda\mu} \alpha^*_{\lambda\mu} Y_{\lambda\mu}(\Omega)\right). \tag{9.144}$$

This is equivalent to the equation

$$V(r_{\Omega}, \Omega) = V_0(r), \tag{9.145}$$

where $V_0(r)$ is the undeformed potential. For any r_{Ω} we can therefore express the actual potential simply by rescaling the coordinate:

$$V(r_{\Omega}, \Omega) = V_0\left[\frac{r_{\Omega}}{1 + \sum_{\lambda\mu} \alpha^*_{\lambda\mu} Y_{\lambda\mu}(\Omega)}\right]. \tag{9.146}$$

For small deformations we can expand and obtain (we replace r_{Ω} by r):

$$V(r, \Omega) = V_0(r) - r\frac{dV_0(r)}{dr} \sum_{\lambda\mu} \alpha^*_{\lambda\mu} Y_{\lambda\mu}(\Omega) + O(\alpha^2). \tag{9.147}$$

The coupling term is therefore of the form:

$$H_{\text{coupl}} = -k(r) \sum_{\lambda\mu} \alpha^*_{\lambda\mu} Y_{\lambda\mu}(\Omega), \tag{9.148}$$

with

$$k(r) = r\frac{dV_0}{dr}. \tag{9.149}$$

After quantizing the surface modes, as in the liquid drop model (Chap. 1), we obtain an expression similar to H_{coupl} of Eq. (9.135). With the assumption that the potential roughly follows the density, we get for the form factor $k(r)$ the same expression for the quadrupole mode (up to a constant factor) within the Tassie (see Sec. 13.3.3) and the liquid drop model, but a different power for higher multipoles. This kind of phenomenological particle vibration coupling vertex has been used in numerous applications over the years for the coupling of a nucleon to either the 2^+ or 3^- surface vibration (see, for instance, the review articles by Arima and Hamamoto

* In this discussion we neglect the difference between RPA and TDA amplitudes.

[AH 71b] and Hamamoto [Ha 74a] and references cited therein, and the conference report on *Problems of Vibrational Nuclei*, by Alaga, Paar, and Sips [APS 75]).

9.3.5 Effective Charges

In this section we apply the boson expansion method to systems with odd particle number to study the concept of the so-called "effective charge." For this purpose we start with the Hamiltonian [(9.131) and (9.135)], make a canonical transformation to RPA bosons (to lowest order in the coupling this is not difficult), and obtain the following expression.

$$H = E_{\mathrm{RPA}} + \sum_k \epsilon_k c_k^+ c_k + \sum_\mu \Omega_\mu O_\mu^+ O_\mu + \sum_{\mu kl} \gamma_{kl}^\mu O_\mu^+ c_k^+ c_l + \text{h.c.}, \tag{9.150}$$

with

$$\gamma_{kl}^\mu = \sum_{mi} \left(X_{mi}^{\mu *} \bar{v}_{mkil} + Y_{mi}^{\mu *} \bar{v}_{ikml} \right). \tag{9.151}$$

(We have dropped the factor $\frac{1}{2}$, since we are going to apply lowest-order perturbation theory; see discussion in footnote on p. 386.) In first-order perturbation theory, the wave function of the odd system is given by:

$$|\tilde{k}) = c_k^+ |0) + \sum_{\mu l} \frac{\gamma_{lk}^\mu}{\epsilon_k - \epsilon_l - \Omega_\mu} O_\mu^+ c_l^+ |0). \tag{9.152}$$

We are interested in the contribution of this "particle–phonon" correction to the matrix element of an electromagnetic multipole operator T. For this purpose, we first have to transform T into the boson representation (9.130). Using the formulas (E.11),

$$T = T^0 + \sum_{mn} T_{mn} c_m^+ c_n - \sum_{ij} T_{ij} c_j c_i^+ + \sum_{mi} (T_{mi} c_m^+ c_i + \text{h.c.}), \tag{9.153}$$

we find, to first order, *

$$T_B = \sum_{mn} T_{mn} c_m^+ c_n - \sum_{ij} T_{ij} c_j c_i^+ + \sum_\mu (T^\mu O_\mu^+ + \text{h.c.}), \tag{9.154}$$

with

$$T^\mu = \sum_{mi} \left(T_{mi} X_{mi}^{\mu *} + T_{im} Y_{mi}^{\mu *} \right). \tag{9.155}$$

We now can calculate the effective matrix element, which is defined by

$$T_{kl}^{\mathrm{eff}} = (\tilde{k} | T_B | \tilde{l}) = T_{kl} + \sum_\mu \left(\frac{\gamma_{kl}^\mu T^{\mu *}}{\epsilon_l - \epsilon_k - \Omega_\mu} + \frac{\gamma_{lk}^{\mu *} T^\mu}{\epsilon_k - \epsilon_l - \Omega_\mu} \right). \tag{9.156}$$

* In the following, we assume that the expectation value T^0 of the core vanishes.

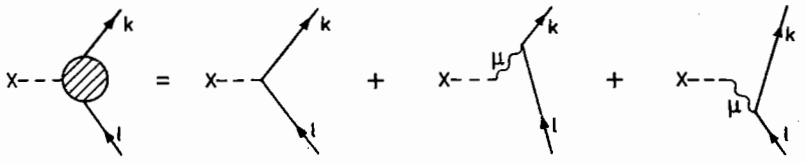


Figure 9.9. Graphical representation of the effective charge.

From this formula we see that the particle vibration coupling changes the bare matrix element T_{kl} to an effective matrix element T_{kl}^{eff}. This is caused by the fact that the external field can excite virtual vibrations of the core which interact with the external particle. If the vibration is very collective, this can give rise to substantial changes in the value of the matrix element. The effective matrix element is graphically represented in Fig. 9.9.

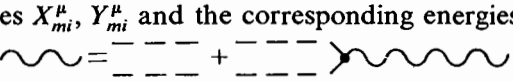
An example of the importance of this process is the effective charges of single-particle states near closed shells. In Section 2.7.2 we saw that it was necessary to introduce an effective charge for protons and neutrons ($e_p^{\text{eff}} \simeq 2$; $e_n^{\text{eff}} \simeq 1$) to reproduce the experimental electric moments and transition probabilities. The reason for this is, of course, the possibility of virtual phonon excitation as shown in Fig. 9.9. Our effective matrix element (9.156) can be written as

$$T_{kl}^{\text{eff}} = T_{kl} + \sum_{mn} e_{klmn}^{\text{pol}} T_{mn}, \tag{9.157}$$

where the polarization charge $e^{\text{pol}} = e^{\text{pol}}(\epsilon_k - \epsilon_l)$ is now an operator which depends on the energy difference $\epsilon_k - \epsilon_l$. Replacing this operator by a constant number corresponds to the concept of the effective charge:

$$e^{\text{eff}} = 1 + e^{\text{pol}}. \tag{9.158}$$

For instance, for the calculation of the effective charge of a quadrupole moment, we must sum up the contributions of all collective 2^+ states of the even core nucleus. In particular, this means that we must also include the giant quadrupole resonance (see Chap. 8). Calculations of this type have been carried out in the Pb region [RBS 73], for example, and the results show good agreement with experiment.

The calculation of the effective matrix elements requires the knowledge of RPA amplitudes X_{mi}^{μ}, Y_{mi}^{μ} and the corresponding energies Ω_{μ}. The RPA equation (8.131)  can be used to derive an equation for T^{eff} directly (see also [Mi 67]):

$$T_{kl}^{\text{eff}}(\omega) = T_{kl} + \sum_{pq} \bar{v}_{kqlp} \frac{n_q - n_p}{\omega - \epsilon_p + \epsilon_q} T_{pq}^{\text{eff}}(\omega). \tag{9.159}$$

9.3.6 Intermediate Coupling and Dyson's Boson Expansion

The perturbative treatment of the particle vibration coupling as described in the preceding sections is, of course, only valid in the extreme weak coupling limit where the off diagonal matrix elements of H_{coupl} are very

small compared to the diagonal ones. As a matter of fact, this situation is not completely fulfilled even in the prototype of weak coupling, namely the septuplet of ^{209}Bi (see discussion below). In order to resolve the problem, we could, of course, diagonalize the Hamiltonian [(9.130), (9.131)] in the basis of the unperturbed one ($H_{\text{coupl}}=0$), but at the same time we should push the boson expansion (9.131) of the Hamiltonian further to be sure that horizontal (see Fig. 9.3) convergence has been achieved. This program has not, however, been carried out so far, and it is rather a different type of boson theory which has been applied mainly to odd nuclei; this is the nuclear field theory (NFT) (see, e.g., [Re 75b, BBB 77] and references cited therein). This theory is *nonlinear*, and it can be shown [IRS 79] that there is a simple mathematical relationship which reduces NFT to the *linear* theory of Dyson's boson expansion (DBE), which has itself also been applied to realistic cases [DJ 73, RS 74b, SWR 76, IRS 80]. Since linear theories are usually simpler to handle than nonlinear ones, we shall present only DBE here. We can also use the formalism which we derived for the even case to a large extent. It is not surprising, that the DBE is also finite in the odd case.

For collective bosons, the fermion pair operators are expressed in the following way (for the non-collective version, see [DJ 73]).

$$\begin{aligned}(b_\mu^+)_B &= B_\mu^+ - \frac{1}{2}\sum_{\nu\rho\sigma}\Gamma^{\nu\rho}_{\mu\sigma}B_\nu^+B_\rho^+B_\sigma - \sum_{kl\nu}\Gamma^{\nu k}_{\mu l}B_\nu^+\beta_k^+\beta_l,\\ (b_\mu)_B &= B_\mu,\\ (a_{\mu\nu})_B &= \sum_{\rho\sigma}\Gamma^{\mu\rho}_{\nu\sigma}B_\rho^+B_\sigma + \sum_{kl}\Gamma^{\mu k}_{\nu l}\beta_k^+\beta_l.\end{aligned} \tag{9.160}$$

Inserting these expressions into the Hamiltonian (9.32) and diagonalizing it in the basis

$$|A+1,\nu)=\sum_p C_p^{(\nu)}\beta_p^+|0)+\sum_{p\mu}C_{p\mu}^{(\nu)}\beta_p^+\tilde{B}_\mu^+|0) \tag{9.161}$$

yields the following matrix equation.

$$\sum_{p'\mu'}\begin{bmatrix}\delta_{pp'}E_p & \frac{1}{2}\gamma^{\mu'*}_{p'p}\\ \frac{1}{2}\gamma^{\mu}_{pp'} & (\Omega_\mu+E_p)\delta_{pp'}\delta_{\mu\mu'}+B_{p\mu p'\mu'}\end{bmatrix}\begin{bmatrix}C_{p'}^{(\nu)}\\ C_{p'\mu'}^{(\nu)}\end{bmatrix}=\omega^{(\nu)}\begin{bmatrix}C_p^{(\nu)}\\ C_{p\mu}^{(\nu)}\end{bmatrix}, \tag{9.162}$$

with $\gamma^\mu_{p'p}$ defined in Eq. (9.132) and

$$B_{p\mu p'\mu'}=-\sum_l(\Omega_{\mu'}-E_l-E_p)C^{\mu'}_{pl}C^{\mu*}_{p'l}. \tag{9.163}$$

In (9.162) we use the quasi-particle TDA equation [(8.10) and (9.70)]. The evaluation of the matrix elements (9.162) is straightforward, and only the treatment of H^{11} [see Eq. (9.32)] is somewhat lengthy. The summation over μ', of course, includes *ph* and *pp* vibrations in the limit of no-pairing correlations. This can easily be worked out with (9.133), (8.201), and (8.202). As can be seen from (9.162), the matrix is, as in the case of even particle number, non-Hermitian. In order to study the significance of the

matrix B, we have to go over to the HF limit, where we can distinguish between *ph* and *pp* vibrations. We then see that B can couple $p-ph$ with $h-pp$ configurations as well as $p-ph$ with $p-ph$ states. According to Fig. 9.8c, the latter case takes care of the Pauli principle. The term $B_{h-pp,\,h-pp}$ is missing because, as is obvious, there is no problem with the Pauli principle.

The meaning of the coefficients $C_{p\mu}^{(\nu)}$ of course, indicates to what extent the collective vibration is built up by the uncorrelated motion of the odd quasi-particle and the collective vibration of the even nucleus (spectroscopic factor). The quasi-particle can be a particle or a hole. The coefficients $C_{p\mu}^{(\nu)}$, therefore, give the probability of, for example, how much the states of the septuplet in ^{209}Bi are built up from a particle coupled to the *ph*-vibrations of ^{208}Pb and how much they are built up from a hole coupled to *pp* vibrations of ^{210}Pb, ^{210}Po, and ^{210}Bi. It should be noted that Eq. (9.162) corresponds to a *nonorthogonal basis*, and therefore only statements about relative probabilities can be made.

The dimension of the matrix (9.162) is, of course, much larger than that for the original fermion space, but we can hope that it can be drastically truncated because of the collectivity of the bosons. Because of the overcompleteness of our basis, spurious solutions are contained in (9.162), but, as in the even case [see Eq. (9.98)], they all lie at the free three-quasiparticle energies, and then are much higher than the interesting collective vibrations. Therefore, for cases where vertical convergence (see Fig. 9.3) has been achieved, these cause no trouble. In this case, all the eigenvalues are also real and the non-Hermiticity of the matrix causes no problem. For cases where we do not have very pronounced collective states and the spurious solutions are therefore intermingled with the physical ones, it is advantageous to transform (9.162) to a Hermitian problem in diagonalizing the norm matrix and eliminating its zero (i.e., practically smaller than a certain value ϵ) eigenvalues [IRS 80, BSL 79]. Taking the square root of the remaining diagonal matrix and inverting yields a Hermitian eigenvalue problem. This corresponds to an orthogonalization of the basis. The details of this procedure are explained in Section 10.2. The dimension of the Hermitian problem can, of course, never become larger than the original fermion space, and therefore no spurious solutions will enter the problem in this way. An orthonormalized basis is also preferable for the calculation of transition probabilities.

In studying these questions it is very instructive to treat an analytically solvable, though academic, model: In the case of no pairing correlations and three particles outside a closed shell core, the vertices γ in (9.162) vanish and we obtain the following simple matrix equation (p, m, n are particles and τ, τ' *pp*-vibrations):

$$\sum_{\tau',\,p'} B_{p\tau p'\tau'} C_{p'\tau'}^{(\nu)} = \left(\omega^{(\nu)} - \Omega_\tau - \epsilon_p\right) C_{p\tau}^{(\nu)}, \tag{9.164}$$

with

$$B_{p\tau p'\tau'} = \sum_n C_{p'n}^{\tau *} (\Omega_{\tau'} - \epsilon_n - \epsilon_p) C_{np}^{\tau'}, \tag{9.165}$$

where C_{mn}^{τ} and Ω_τ are solutions of a *pp*-TDA calculation. As our model, we take

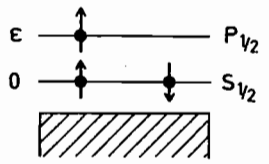


Figure 9.10. The three-particle configuration of the model.

the case of three valence nucleons in an $s_{1/2}$ and $p_{1/2}$ level (see Fig. 9.10). There exists only one state with negative parity and spin up:

$$|\alpha\rangle = a_s^+ {}_{\frac{1}{2}} a_{s-\frac{1}{2}}^+ a_p^+ {}_{\frac{1}{2}} |D\rangle.$$

The energy of this state can be readily calculated to be

$$\begin{aligned}\langle\alpha|H|\alpha\rangle - \langle D|H|D\rangle &= \epsilon + v_{s\frac{1}{2}s-\frac{1}{2},s\frac{1}{2}s-\frac{1}{2}} + v_{s\frac{1}{2}p\frac{1}{2}s\frac{1}{2}p\frac{1}{2}} + v_{s-\frac{1}{2}p\frac{1}{2}s-\frac{1}{2}p\frac{1}{2}} \\ &= \epsilon + G + F + H.\end{aligned}$$

We now have to calculate the two-particle states; we suppose that only the diagonal matrix elements are nonzero and obtain the following five relevant τ states:

$$\begin{aligned}C_s^{\tau_1}{}_{\frac{1}{2}p\frac{1}{2}} &= 1, & \Omega_{\tau_1} &= \epsilon + F; \\ C_{s-\frac{1}{2}p\frac{1}{2}}^{\tau_2} &= 1, & \Omega_{\tau_2} &= \epsilon + H; \\ C_s^{\tau_3}{}_{\frac{1}{2}s-\frac{1}{2}} &= 1, & \Omega_{\tau_3} &= G; \\ C_s^{\tau_4}{}_{\frac{1}{2}p-\frac{1}{2}} &= 1, & \Omega_{\tau_4} &= \epsilon + v_{s\frac{1}{2}p-\frac{1}{2}s\frac{1}{2}p-\frac{1}{2}} := \epsilon + I; \\ C_p^{\tau_5}{}_{\frac{1}{2}p-\frac{1}{2}} &= 1, & \Omega_{\tau_5} &= 2\epsilon + v_{p\frac{1}{2}p-\frac{1}{2}p\frac{1}{2}p-\frac{1}{2}} := 2\epsilon + J.\end{aligned}$$

The corresponding five $p\tau$ configurations are displayed in Fig. 9.11.

We can clearly see how neglecting the Pauli principle in the basis states enlarges the dimension of the space from one to five, that is, the space is now overcomplete. Let us now calculate the corresponding matrix (we abbreviate the different configurations $p\tau$ by I, II,...):

$$\begin{aligned}B_{1\,\mathrm{I}} &= \sum_n C_{s-\frac{1}{2}n}^{\tau_1}(\epsilon + F - \epsilon_n - 0) C_{ns-\frac{1}{2}}^{\tau_1} = 0, \\ B_{1\,\mathrm{II}} &= \sum_n C_s^{\tau_1}{}_{\frac{1}{2}n}(\epsilon + H - \epsilon_n - 0) C_{ns-\frac{1}{2}}^{\tau_2} = -H, \\ B_{1\,\mathrm{III}} &= -G, \\ B_{1\,\mathrm{IV}} &= B_{1\,\mathrm{V}} = 0.\end{aligned}$$

Continuing in this way for the entire matrix, we arrive at the following secular equation for the eigenvalues of Eq. (9.164).

$$\begin{vmatrix} \epsilon + F - \omega^{(\alpha)} & -H & -G & 0 & 0 \\ -F & \epsilon + H - \omega^{(\alpha)} & G & 0 & 0 \\ -F & H & \epsilon + G - \omega^{(\alpha)} & 0 & 0 \\ 0 & 0 & 0 & \epsilon - \omega^{(\alpha)} & 0 \\ 0 & 0 & 0 & 0 & 2\epsilon - \omega^{(\alpha)} \end{vmatrix} = 0.$$

It is obvious that the corresponding matrix is non-Hermitian, and it can be readily checked that the eigenvalues are:

$$\omega^{(1)} = \omega^{(2)} = \omega^{(3)} = \epsilon; \qquad \omega^{(4)} = 2\epsilon; \qquad \omega^{(5)} = \epsilon + F + G + H.$$

We thus get back the physical eigenvalue and four unphysical ones which lie at the

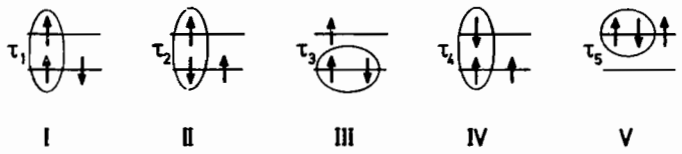


Figure 9.11. The five possible particle boson configurations.

unperturbed particle energies. This, of course, is the same as what we have already discussed in the case of the DBE for even systems. The unphysical states all lie at the unperturbed particle (hole) energies. Of course, our model has only had a pedagogic purpose in order to explain the principle involved. In realistic cases, where there may be quite collective states of the even system, the actual matrix to be diagonalized is much smaller than the one in the fermion space.

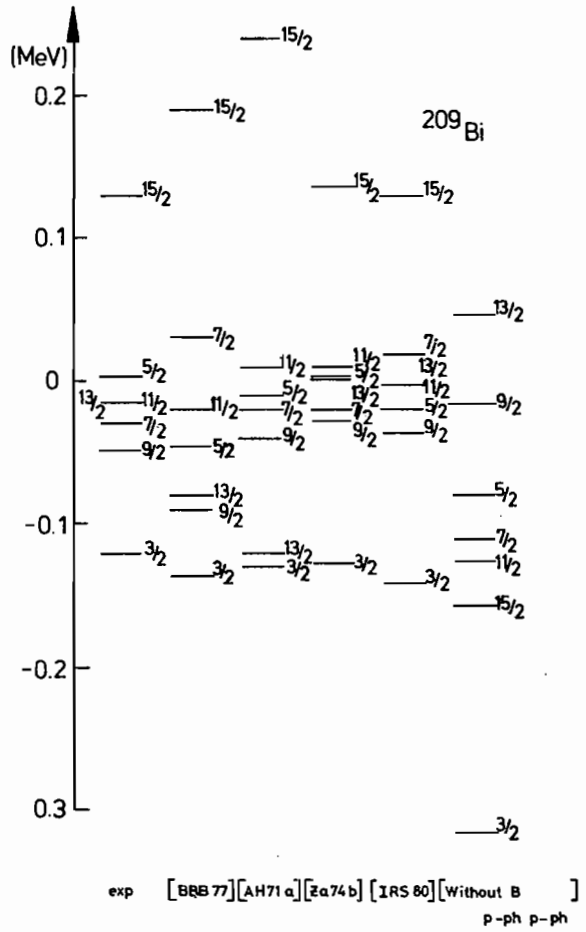


Figure 9.12. The spectrum of the particle vibration coupling states (septuplet) in ^{208}Bi as given by experiment and the calculation by different authors. For the last column, the matrix $B_{p-ph,\ p-ph}$ of (9.162) was omitted in order to demonstrate the strong influence of the Pauli principle. The zero energy line corresponds to the energy of the 3^- state at 2.614 MeV in ^{208}Pb.

As an example, we show in Fig. 9.12 the septuplet of ^{209}Bi, which has been calculated solving Eq. (9.162) by enlarging the dimension successively until vertical convergence (see Fig. 9.3) is achieved. In Fig. 9.12 the spectra calculated with nuclear field theory are also given [BBB 77] together with calculations of other authors [AH 71a, Za 74b] using different approaches.

In Table 9.2 we give certain components $C_{p\mu}^{(\nu)}$ (spectroscopic factors) in order to see their relative importance and in Fig. 9.12 we show how relevant it is to treat the Pauli principle correctly. This can be checked by switching off the $p-ph$, $p-ph$ part of the matrix B of Eq. (9.162). The remaining discrepancies with the experimental spectrum can be due to several things. One obvious feature which has been omitted is the fact that the $2p-1h$ components are the main components of the septuplet and we treated the weak coupling to the $1p$ space but neglected the coupling to the $3p-2h$ configuration; by symmetry reasons the latter should then also be included, a task much harder to realize in practice.

Table 9.2. Wave functions of several particle vibrational coupling states in the septuplet of ^{209}Bi given by the spectroscopic factors $C_{p\mu}$. The largest component $\pi h_{9/2}\otimes^{208}\mathrm{Pb}(3^-)$ is normalized to unity. (From [IRS 80].)

p μ	3/2	5/2	7/2
Pure single particle	0.005	0.001	0.026
$\pi h_{9/2}\otimes^{208}\mathrm{Pb}\ (3^-)$	1.0	1.0	1.0
$\pi h_{9/2}\otimes^{208}\mathrm{Pb}\ (6_4^-)$	−0.385	0.266	−0.233
$\nu g_{9/2}\otimes^{208}\mathrm{Bi}\ (6_2^+)$	−0.406	0.356	−0.320
$\nu i_{11/2}\otimes^{208}\mathrm{Bi}\ (6_1^+)$	0.100	−0.155	0.211
$\nu i_{11/2}\otimes^{208}\mathrm{Bi}\ (7_1^+)$	−0.318	0.282	−0.232
$(\pi d_{3/2})^{-1}\otimes^{210}\mathrm{Po}\ (0^+)$	0.326	—	—
$(\pi d_{3/2})^{-1}\otimes^{210}\mathrm{Po}\ (2^+)$	0.233	0.231	0.095
$(\pi d_{3/2})^{-1}\otimes^{210}\mathrm{Po}\ (4^+)$	—	0.200	0.282
$(\nu p_{3/2})^{-1}\otimes^{210}\mathrm{Bi}\ (1^-)$	0.286	−0.128	—
$(\nu p_{3/2})^{-1}\otimes^{210}\mathrm{Bi}\ (2^-)$	0.225	0.205	0.090
$(\nu p_{3/2})^{-1}\otimes^{210}\mathrm{Bi}\ (3^-)$	0.151	0.234	0.191
$(\nu p_{3/2})^{-1}\otimes^{210}\mathrm{Bi}\ (4^-)$	—	−0.181	−0.255
$(\nu p_{3/2})^{-1}\otimes^{210}\mathrm{Bi}\ (5^-)$	—	—	−0.235

9.3.7 Other Particle Vibration Coupling Calculations

As we said at the beginning, there are many applications of different weak coupling formalisms all of which we do not wish to present here. For instance, we have not treated any of the quasi-particle–phonon coupling theories [KS 63, DJ 73] for spherical open-shell nuclei, where pairing correlations have to be included. One of the most advanced theories in this respect which is, however, beyond the scope of this book is the one set up by Marumori and coworkers [KMM 72, 75(a+b)]. This theory is able to treat

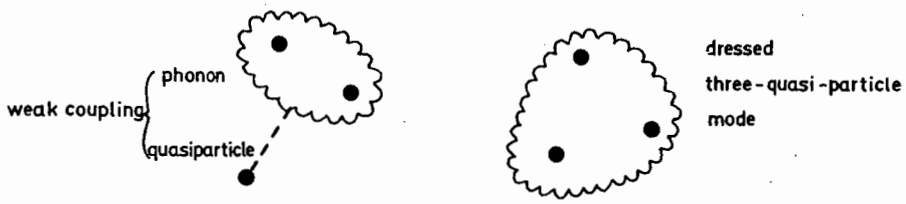


Figure 9.13. Schematic representation of a quasi-particle phonon and a dressed three-quasi-particle state.

not only the weak but also the strong coupling case where three quasi-particles are strongly interwoven (Fig. 9.13) (of course, this is also the case for the DBE) and takes into account ground state correlations as well as, for instance, a proper elimination of the spurious components due to particle number violation of the BCS theory. A nice application of this theory, in which can be seen a gradual transition from weak to strong coupling is the $h_{11/2}$ odd neutron coupled to the 2^+ phonon state for different Xe and Te isotopes, shown in Fig. 9.14. In Xe we see a gradual increase of the splitting in going from ^{133}Xe to ^{127}Xe—that is, towards the middle of the $h_{11/2}$ shell for ^{127}Xe the splitting has become so strong that the lowest member—that is, the $9/2$ (correlated three quasi-particles) state is lower than the single-particle state $11/2$ of the odd neutron. Such states are usually called *abnormal coupling states*, and are due to two effects:

(i) a geometrical reason, which is that in a j-particle 2^+-phonon coupling the $(j-1)$ state is always the lowest member of the multiplet (see [KMM 75a]); and
(ii) the increase of the ground state (BCS) correlations as the middle of the shell is approached. It can be shown that graph c in Fig. 9.8 then becomes very important.

The coupling is less pronounced in the Te isotopes, and no abnormal coupling occurs. The experimental trends are nicely reproduced by the theory.

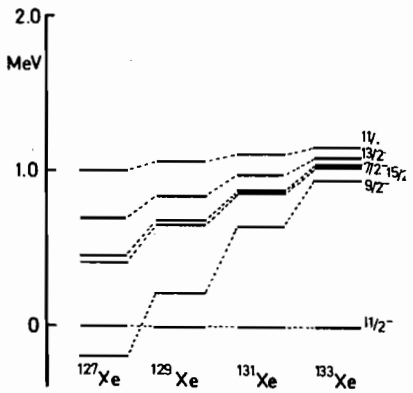


Figure 9.14. Energy systematics of particle vibrational coupling states for a series of Xe-isotopes. (From [KMM 75a].)

9.3.8 Weak Coupling in Even Systems

There are many other different kinds of weak coupling situations. A weak coupling need not necessarily occur only in odd mass nuclei—there are also weak coupling situations in even nuclei. For example, the two 3^--phonon quadruplet in ^{208}Pb ($3^-\otimes 3^- \Rightarrow 0^+, 2^+, 4^+, 6^+$; only the 0^+ and 2^+ have been experimentally identified so far) is likely to be of the weak coupling type [BBN 74a, Sch 76]. On the other hand, it is preferable to treat such a problem with a formalism able to describe weak, intermediate, and strong coupling situations at the same time and we have seen above that the Dyson boson expansion (DBE) is such a formalism for the case of magic or close to magic nuclei. Treating the coupling of one- and two-phonon states, we can write for the wave function:

$$|\tilde{\nu}) = \sum_{\mu} C_{\mu}^{(\nu)} \tilde{B}_{\mu}^{+} |0) + \sum_{\mu_1 \mu_2} C_{\mu_1 \mu_2}^{(\nu)} \tilde{B}_{\mu_1}^{+} \tilde{B}_{\mu_2}^{+} |0), \tag{9.166}$$

where the μ sums, of course, again include *ph* and *pp* vibrations. The corresponding DBE-matrix has the form

$$\begin{bmatrix} \Omega_{\mu}\delta_{\mu\mu'} & -\frac{1}{2}\sum_{kl} \gamma_{kl}^{\mu_1'} \Gamma_{\mu' l}^{\mu_2' k} - (\mu_1' \leftrightarrow \mu_2') \\ -\frac{1}{2}\sum_{kl} \gamma_{kl}^{\mu_1} \Gamma_{\mu' l}^{\mu_2 k} - (\mu_1 \leftrightarrow \mu_2) & (\Omega_{\mu_1} + \Omega_{\mu_2})\delta_{\mu_1\mu_1'}\delta_{\mu_2\mu_2'} + W_{\mu_1\mu_2\mu_1'\mu_2'} \end{bmatrix}, \tag{9.167}$$

with γ_{kl}^{μ} given in Eq. (9.132), $\Gamma_{\mu' l}^{\mu k}$ in Eq. (9.29), and $W_{\mu_1\mu_2\mu_1'\mu_2'}$ in Eq. (9.72). This equation has been numerically solved for the $3^-\otimes 3^-$ quadruplet in ^{208}Pb [Sch 76]. Convergence in the vertical direction (see Fig. 9.3) has been achieved for about 10–20 configurations, and the splitting of the quadruplet found to be of the order of 100 keV, indicating quite a weak coupling situation.

In the case of superfluid nuclei, a more elaborate theory has to be set up due to particle number violation. This has been done by the Marumori group [IMS 76]. In this way, it has been possible to explain, for example, the anomalously low position of the second 0^+ state in some Ge isotopes which turns out to be due to a very anharmonic interplay between the pairing vibration and the 0^+ member of a $(2^+\otimes 2^+)$ quadruplet. The fact that these Ge isotopes are very "soft" with respect to pair vibration can be understood from the very low-level density ($j=1/2$ states) below the fermi surface and the very high-level density just above the fermi level ($j=9/2$) in these nuclei. As discussed in Chapters 6 and 7, pairing correlations are favored by a high-level density, and the nucleons therefore have a strong tendency to exploit the phase space offered to them just above the fermi level. This makes the nucleus very anharmonic in the pairing mode. These considerations conclude our studies about anharmonic effects in nuclei, in which we have seen that the boson descriptions are adequate to describe even and odd nuclei. Further numerical applications have, however, to be carried out to justify the method in all its aspects from a practical point of view.